UNIVERSITY OF GOTHENBURG

DEPARTMENT OF BIOLOGICAL AND ENVIRONMENTAL SCIENCES

A Comparative Study of Air Quality and Urban Development in Sri Lankan Cities (1980-2024)

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Abstract

The aim of this study is to investigate the long -term relationship between urbanization and air quality in four Sri Lankan cities, Colombo, Kandy, Anuradhapura, and Jaffna from 1980 to 2024. As rapid urbanization and economic growth change Sri Lanka's cities through accelerated urbanization, fine particulate matter ($PM_{2.5}$) has emerged as a significant environmental and public health concern. This study employs satellite-derived data from NASA's MERRA-2 and Washington University and ground-based measurements at the U.S. Embassy in Colombo to assess $PM_{2.5}$ and component concentrations, including black carbon (BC), organic carbon (OC), sulfate (SO_4^{2-}), dust, and sea salt (SS).

Urbanization metrics of population growth, land use change, and household energy consumption were assessed against $PM_{2.5}$ concentrations through time series and comparative analyses. The findings exhibit distinct spatial and temporal pollution trends: while cities like Colombo and Anuradhapura experience stable trends of pollutants, Kandy experiences noticeable increases in $PM_{2.5}$, BC, and SO_4 likely to be influenced by topography and urbanization. Conversely, due to a lack of extensive data availability, population trends in Jaffna were not able to be reasonably evaluated. Seasonal and diurnal analysis also refers to the influence of climatic features like monsoons in defining air quality.

This study contributes to the limited body of literature on long-term air quality monitoring in Sri Lanka and makes recommendations for environmental policy and sustainable urban planning. The study also underscores the value of more data collection and integration to guide air quality management in the developing world.

1. Introduction

Urbanization is one of the key factors for economic and social change in developing nations. As a result of rapid urbanization, economic growth, and development seen during the past few decades, air pollution levels have risen in urban areas of Sri Lanka. Among them, fine particulate matter, or $PM_{2.5}$, has emerged as the most significant air pollutant owing to its detrimental effect on the environment and human health.

$PM_{2.5}$ is made up of suspended particulate matter with diameters of less than 2.5 micrometers. The particles comprise black carbon (BC), organic carbon (OC), sulfates, dust, and sea salt. Both natural and anthropogenic sources are responsible for the emission of these particles into the atmosphere, magnifying their effects on both environmental aspects and human health. $PM_{2.5}$ further influences climate change by interaction with solar radiation and cloud formation as well as visibility reduction, thus impacting day-to-day activities.

The principal sources of air pollution in Sri Lanka include vehicular emissions, industries, and power generation. These sources are closely linked to industrialization and urban development in the country. The increased population, vehicles, and infrastructure in cities have led to a rise in air pollution levels. Furthermore, the physical nature of cities such as Kandy is accountable for elevated pollutant levels. This has posed a question about the sustainability of urban development and the health impacts on urban residents in the long term.

The lack of long-term studies examining the relationship between air quality and urban development using satellite-derived data in Sri Lankan cities represents significant research gap. The present thesis intends to bridge this gap by examining the effect of urban extension on $PM_{2.5}$ concentrations in chosen cities of Sri Lanka for the period 1980-2024. In the present study, an ensemble of satellite-retrieved datasets such as NASA's MERRA-2 and Washington University data and ground measurements of air quality data provided by the US Embassy will be utilized to enhance the authenticity of the data collected. In addition, socio-economic indicators related to urbanization and population growth were analyzed to assess their correlation with air quality trends.

Using time series and comparative analysis methods, this research will enhance the knowledge on interlinkages of urban development and $PM_{2.5}$ air pollution and its critical constituents. Results are expected to guide urban planning, environmental management, and public health policy in Sri Lanka, and potentially other developing nations that experience similar issues.

2. Background Information

2.1 Particulate Matter 2.5 ($PM_{2.5}$), and its components.

2.1.1 Particulate Matter 2.5 ($PM_{2.5}$)

Airborne particulate matter consists of a combination of solid particles and liquid droplets, including sulfate (SO_4^{2-}), nitrate (NO_3^-), ammonium (NH_4^+), Black carbon (BC), Organic carbon (OC), dust and sea spray, that vary in size and chemical composition (Elmarakby & Elkadi, 2024; Schwartz et al., 1996). These particulate particles are suspended in the air for a long time due to their light weight (CAICE, 2025). The Environmental Protection Agency (EPA) categorizes particle pollutants into two primary groups by considering their diameter (Elmarakby & Elkadi, 2024). A diameter of particulate matter between 2.5 and 10 μm is known as a coarse particle (PM_{10}); less than 2.5 μm is considered a fine particle, which is one of the key indicators of air pollution (Ali et al., 2022; Elmarakby & Elkadi, 2024; Schwartz et al., 1996). Spatial concentrations in $\mu g\ m^{-3}$ are generally used to measure the distribution of PM pollution(Akalanka et al., 2024; Zu & Gao, 2024).

According to Ali et al. (2022), direct emissions of PM occur in the atmosphere through two types of sources, which are natural emission sources and anthropogenic sources. Biogenic emissions, dust, soil, and sea salt contribute to natural particle emissions, while fossil fuel combustion, construction activities, industrial activities, transportation, and deforestation generate $PM_{2.5}$ as a significant human influence on emissions. Chemical reactions of gases such as NO_x, SO_2, NH_3, and volatile organic compounds (VOCs) in the atmosphere, released from combustion, lead to the formation of secondary aerosol particles, e.g. sulfate particles (Ali et al., 2022; Elmarakby & Elkadi, 2024; Schwartz et al., 1996).

Organic and inorganic substances such as sulfate (SO_4^{2-}), nitrate (NO_3^-), ammonium (NH_4^+), Black carbon (BC), and Organic carbon (OC) are several components of $PM_{2.5}$ (Zhang et al., 2025). Sulfate mainly comes from industries, transportation, and power plants while combusting fossil fuels. Black carbon is added to the atmosphere due to incomplete fossil fuel combustion and the burning of biomass and wood for industrial and household activities. Vehicles and industries are some of the sources that release OC into the air (Zhang et al., 2025).

2.1.2 Sulfate (SO_4^{2-})

Sulfate (SO_4^{2-}) particle that is produced by the combustion of fossil fuels and biomass. The gaseous form of SO_2 in the atmosphere is transformed into a sulfate particle through various chemical processes (Khan et al., 2010; Kristjánsson, 2002). Owing to the changes in emission regulations and improvements in combustion technology, the sulfate concentration has been reported to decrease by approximately 59% in the United States from 1979 to 2002 (Khan et al., 2010).

The factors that contribute to sulfate particles are fossil fuel burning, oil refining activities, heavy traffic, waste incineration, and biomass burning in Sri Lanka

2.1.3 Black Carbon (BC)

Black carbon (BC) or soot is one of the aerosol particles that forms due to the incomplete combustion of fossil fuels or biomass (IPCC, 2007). According to the IPCC's Fourth Assessment Report, (2007), Black carbon emissions in the UK, Germany, the former Soviet Union, and the USA were reported to have decreased, while increasing in Asia and China over the period from 1950 to 2000.

2.1.4 Organic Carbon (OC)

Organic aerosols consist of a diverse range of carbon-based chemical compounds. Organic carbon emissions occur due to the burning of fossil fuels and biofuels, as well as open burning, which are anthropogenic OC emissions. Wildfires and emissions from vegetation are natural sources. The chemical reaction of primary volatile organic substances present in the atmosphere creates secondary OC (IPCC, 2007). According to the IPCC's fourth assessment report, (2007), the estimated organic carbon emissions from fossil and biofuels increased by a factor of three over the period 1870 to 2000. The primary emission sources of organic aerosol in urban areas are road traffic, industrial processes, waste incineration, wastewater treatment processes and domestic heating (Abdul-Razzak, 2012).

2.1.5 Mineral dust

The primary source of dust production is located in arid and semiarid regions worldwide, which account for approximately one-third of the world's land area (NASA, 2025); Seinfeld & Pandis, 2016). Dust is emitted into the air by natural and human activities. The blowing wind over dry soil naturally adds dust; on the other hand, deforestation, overgrazing, and the construction industry are sources of anthropogenic dust influence (Elmarakby & Elkadi, 2024; Fang et al., 2024; NASA, 2025).

2.1.6 Sea Salt

Sea salt, which is one of the major PM of natural origin, is the primary component in sea spray. Additionally, organic matter, including dissolved organic carbon, bacteria, phytoplankton, and microalgae, is also included in it (Lin et al.; NASA, 2025) Sea salt is generally produced by the action of wind on the ocean. The sea waves are created on the ocean surface by wind force. Breaking sea waves produce air bubbles at the sea surface, creating whitecaps and bursts. This process releases seawater drops into the atmosphere. Wind speed, ambient relative humidity, and sea surface temperature cause on formation of sea salt (Hu et al., 2024; Lewis & Schwartz, 2004; NASA, 2025; Reich et al., n.d.).

2.2 Impact on the Environment

Increasing PM levels cause serious environmental problems. It reduces air quality and atmospheric visibility. Haze pollution in Beijing-Tianjin-Harbin region in China in 2013 severely affected the normal life of more than 80 million people due to flight delays and highway closures, in addition to respiratory diseases (Wu et al., 2020). Furthermore, PM depositions can have adverse environmental impacts, such as damage to vegetation, pollution of soil and water resources, and harm to wildlife (Elmarakby & Elkadi, 2024).

2.3 Impact on Human Health

The entire world is concerned about and monitoring the air pollutant $PM_{2.5}$ due to its adverse effects on human health (Dhammapala et al., 2022). This particle can enter the deeper levels of the lungs and penetrate the bloodstream. Inhaling fine particles can cause cardiovascular issues, including heart attacks and strokes, as well as other cardiovascular diseases. Furthermore, it increases the risk of cancer (Ameri et al., 2023). The World Health Organization (WHO) indicates that ischemic heart disease (IHD) stands as the leading cause of premature mortality in Europe, with approximately 48 % of this mortality associated with $PM_{2.5}$ (Elmarakby & Elkadi, 2024).

Exposure to $PM_{2.5}$ for a long period causes a decrease in lung function and increases the risk of chronic obstructive pulmonary disease. Decreasing air quality levels adversely affect more than 99% of the world's population. This poor air quality is directly related to more than seven million premature deaths each year globally (Hsu et al., 2025). The estimated deaths due to exposure to $PM_{2.5}$ were 851,660 in China in 2017 (Zhang et al., 2025). According to the World Health Organization (WHO), Sri Lanka experiences approximately 4,300 premature deaths annually due to indoor air pollution, while 1000 deaths result from outdoor air pollution (Shelton et al., 2022).

2.4 Urbanization, Economic growth, population, and Rising Pollution Levels

According to Akalanka et al. **(**2024), the term "urbanization" describes the process by which rural areas transform into urban areas, marked by population migration and city expansion. As a result of this process, more than half of the population resides in major cities today, contributing to a rapidly growing urban population (Relvas et al., 2025). Owing to urbanization, the population growth in major cities has increased. This population growth leads to increased human activity in cities, such as increased daily travel and vehicle use. As a result, the air pollution level increases in the atmosphere (Wang et al., 2025).

Rapid urbanization in developing countries like Sri Lanka is identified as one of the significant factors in increasing air pollution. Owing to the population growth and breakthrough of industrialization**,** the influence of the acceleration of fuel combustion, urban constructions, and traffic emissions leads to high $PM_{2.5}$ concentrations in the atmosphere (Manju et al., 2018; Shelton et al., 2022). In some cases, traffic contributes up to 37% of the $PM_{2.5}$ in the air (Zalakeviciute et al., 2018).

Research conducted in China has found that economic development level, urbanization level, population size, energy intensity, coal combustion, and increased use of private vehicles exacerbate the $PM_{2.5}$ pollutant level (Dong et al., 2018).

A breakthrough in the transport sector due to increasing economic activities has resulted in ameliorating air pollution in major cities. The motor vehicle population more than doubled between 1992 and 2012 in Sri Lanka. In 1996, the primary source of electricity generation was hydropower, and it was more than 90%. Today, the hydro power ratio has gone down to 40 - 50%, and the rest of the power is generated by fossil fuel burning (Ileperuma, 2015).

2.5 Weather parameters, topography, and PM variation

Studies have shown that the shape of the land (topography) and weather parameters such as wind speed, temperature, and precipitation influence the accumulation, movement, and dispersion of $PM_{2.5}$. Temperature, wind and rain aid to reduce $PM_{2.5}$ by spreading it out or washing it out of the air (Manju et al., 2018; Wu et al., 2020; Zalakeviciute et al., 2018).

According to many studies, high humidity and high elevation lead to an increase in particle pollution. At high humidity, the air intake of a vehicle's engine is cooled down by the moist air, which lowers air density and reduces oxygen, causing less fuel to burn completely. Therefore, this leads to more soot, unburned fuel, and hydrocarbons (Zalakeviciute et al., 2018).

The air is cooler and thinner at higher elevations, which leads to more particulate matter due to less oxygen for the combustion of vehicles' engines. A study shows that PM emissions from diesel engines increased by 142% at 1000 meters and by 203% at 3200 meters compared to sea level (Zalakeviciute et al., 2018)

2.6 Study area: Background of Sri Lanka

Figure 1: A. Districts of Sri Lanka

Sri Lanka is an island surrounded by the Indian Ocean and located southeast of India (Gopalakrishnan et al., 2020). The country has diverse topography, such as coastal lowlands, central highlands, and intermediate plains. The highest point is Piduruthalagala at 2524 m above sea level. It is divided into 25 districts, and the total population of Sri Lanka was over 22 million by the year 2024 (Central Intelligence Agency, 2025).

Sri Lanka shifted from a socialist orientation to an open economy with foreign investment in 1977. After this change, the industrial sector in the Sri Lankan economy surpasses the agriculture sector. After this transition, industries such as the apparel and textile sector, the transport sector, and infrastructure development, such as roads and the construction sector, rapidly increased in recent decades (Embuldeniya, 2015).

2.6.1 Climate

Figure 2: Climate Zones of Sri Lanka (Gunathilake, 2020)

Sri Lanka belongs to the tropical climate zone, and it is divided into four climate zones: dry, wet, intermediate, and arid based on rainfall. The dry zone is in the northern and eastern parts of Sri Lanka. Anuradhapura and Jaffna belong to this zone. The wet zone covers southwest, Colombo, and Kandy, located in this zone. The intermediate zone is located between the wet and dry zones. The arid zone is part of the northern and southeastern region with very low rainfall.

2.6.2 Monsoon seasons

The four monsoons: Northeast monsoon, first intermonsoon, Southwest monsoon, and second intermonsoon bring rainfalls to Sri Lanka. Northeast monsoon, locally known as “Maha season “, brings rain from December to February. The northeasterly trade winds blowing from the Indian subcontinent towards the Indian Ocean through the Bay of Bengal bring rain to the northern, eastern, and southeastern regions of Sri Lanka. The first intermonsoon brings rain from March to April. In this period, the rainfall is unevenly distributed, providing short-duration high-intensity rainfall with lightning in some regions while others remain dry (Department of Meteorology, Sri Lanka, n.d.).

The southwest monsoon, also known as “Yala” season locally, brings rain from May to September. This monsoon provides over 60% of the annual rainfall in the west zone in Sri Lanka. The wind blows from the Indian Ocean, bringing most air masses. The heavy rain occurs in Colombo and Kandy during this monsoon period. The second intermonsoon rainfall occurs during October and November. This monsoon season provides short-duration, high-intensity rainfall to most parts of Sri Lanka (Department of Meteorology, Sri Lanka, n.d.).

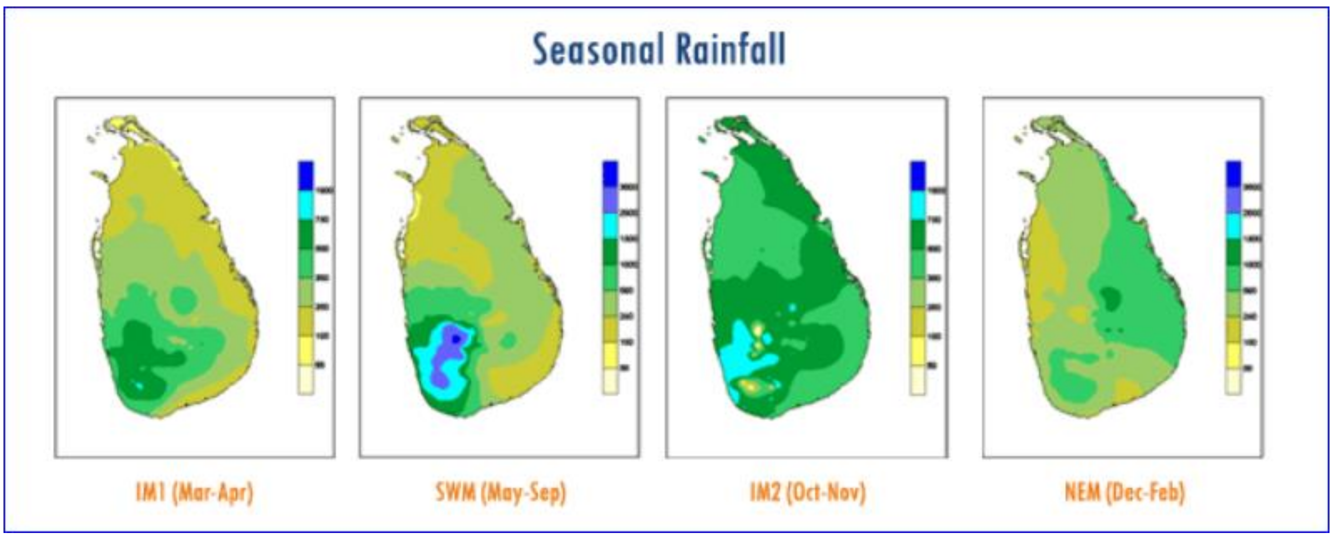

Figure 3:Seasonal rainfall in Sri Lanka (Department of Meteorology, Sri Lanka, n.d.)

2.6.3 Colombo

Colombo, located between 6°55′ and 6°58′ North latitude and 79°50′ and 79°54′ East longitude, is situated on the West Coast of the country, and is the capital of Sri Lanka, bordered by the Indian Ocean. It is a flat terrain city with an elevation from 0 to 10 m above sea level. In addition to that, it is the largest city and commercial hub. The harbor, industrial zone, and energy power plants are also located around the city. The city's roads are mostly congested with private and public vehicles (Seneviratne et al., 2011). After the civil war in 2009, Colombo underwent rapid development, and people migrated to Colombo for work (Wijayamuni et al., 2023).

Colombo is a hot, humid, and tropical city. Southwest (from late May to late September) and Northeast (from late November to mid-February) monsoons affect these climate changes. The average temperature in Colombo varies from 26^{0}C to 31^{0}C. Relative Humidity ranges from 70% during the day to 90% during the night. Higher Rainfall is received during monsoon periods, and it exceeds 2500mm per year (Johansson et al.; 2006, Udayanga et al.; 2020).

2.6.4 Kandy

Kandy belongs to the Central Province and is in the center of the country, approximately 465m above mean sea level, in a valley area, and surrounded by a high mountain range. Kandy is 116 km away from Colombo city. The United Nations Educational, Scientific and Cultural Organization (UNESCO) declared Kandy a World Heritage Site (Dissanayake et al., 2019, 2020). According to the topography of Kandy, the city's road network is narrow and limited in space, making it difficult to expand as the vehicle population grows. As a result**,** Kandy frequently experiences traffic congestion, especially in peak hours (Priyankara et al., 2021).

Kandy experiences approximately 2085 mm of long-term average rainfall and 52-398 mm of monthly rainfall. The average temperature of Kandy City ranges from 20^{0}C to 22^{0}C throughout the year, and its mean daytime temperature ranges from 28^{0}C to 32^{0}C, and the dry period is from January to April, and the monsoon season brings rain to Kandy City. The relative humidity ranges from 63% to 83% (Dissanayake et al., 2020).

2.6.5 Anuradhapura

The Anuradhapura is located in the North Central Province of Sri Lanka. It is important historically, culturally, and agriculturally. Anuradhapura is a rural city situated in flat terrain (Janani et al., 2024).Agriculture is the main industry in this district, and it is considered the primary contributor to paddy production. Therefore, it is known as the heart of Sri Lanka's rice Bowl. Farmers in Anuradhapura engage in Paddy cultivation in two seasons, from September to March and from May to August. (Janani et al., 2024). The main livelihood of district dwellers is agriculture, and others engage in the fishing industry, business, industries, and services sectors (Janani et al., 2024).

It belongs to the dry zone of Sri Lanka. The annual rainfall, which is mainly received from the northeast monsoon, varies between 1250 and 2000 mm in Anuradhapura. The average annual temperature is 28.5^{0}C, and it experiences a long period of drought (Janani et al., 2024).

2.6.6 Jaffna

Jaffna, which has 160 km of coastline and is almost completely surrounded by the Ocean, is located in the northern part of Sri Lanka. Jaffna has flat terrain, and there is no area higher than 15 m above mean sea level (Gopalakrishnan et al., 2020). An ongoing process of Rehabilitation and reconstruction of the Jaffna region has been conducted since the end of the civil war in 2009. 65% of the Jaffna population engages with agriculture as their main livelihood (Gopalakrishnan et al., 2020).

2.7 Policy and Urban Planning

The government of Sri Lanka has taken action to reduce air pollution by implementing the national policy on urban air quality management in 2000. The aims of this policy are to maintain good air quality, prevent air pollutant-related diseases, and reduce national health expenditures in Sri Lanka (Shelton et al., 2022). In addition to that, the Sri Lankan government introduced several measures, which further strengthened the National Environmental (Ambient Air Quality) Regulations of 1994, to improve the air quality. They are introducing low-percentage sulfur diesel in January 2003, prohibiting 2-stroke three-wheelers in 2008, and initiating vehicular emission tests in 2008 (National Environmental Act, 1994, Shelton et al., 2022).

According to the National Ambient Air Quality Standards (NAAQS) in 1994, which were revised in 2008, there are six primary air pollutants: carbon monoxide (CO), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), ozone (O_3), PM_{10}, and $PM_{2.5}$, considered as critical air pollutants (Ileperuma,2020).

3. Methods

3.1 Study Area

The study focused on the four main urban cities in Sri Lanka: Colombo, Kandy, Anuradhapura, and Jaffna. The criteria for selecting these cities were based on their urban development characteristics, topography, and climate, which influence air quality in distinct ways.

Table 1:Longitude and Latitude of study areas

City	Longitude (E)	Latitude (N)
Colombo	79.8612	6.9271
Kandy	80.6337	7.2906
Anuradhapura	80.4037	8.3114
Jaffna	80.0255	9.6615

Colombo is situated on the country’s west coast along the Indian Ocean. It is the capital city of Sri Lanka and serves as the country’s industrial and commercial hub. It is the most urbanized city in Sri Lanka, with high population, industrial activities, and significant traffic congestion contributing to its selection as a central city. Kandy is located in the central highlands of Sri Lanka, surrounded by mountain ranges such as Knuckles and Hanthana. The city sits at an elevation of approximately 500 m above sea level and is 116 km away from Colombo. Aside from its population, industrial activity, traffic, topography is the main reason for selecting this city to analyze the increase in pollutants with urbanization parameters. The Anuradhapura district is situated in the North Central lowlands of Sri Lanka, where agriculture serves as the primary livelihood for many residents. Jaffna is located in the country’s northern part, near the Indian peninsula. The area's topography is generally flat, with a median elevation of 2.72 m. The majority of the population's livelihood is primarily based on agriculture. Differences between each sector of city, aiding to better understanding of pollutant variation with urbanization and its link to rising pollutants.

3.2 Type of data source

Multiple data sources were utilized in this study to gather information on air quality and urban development indicators. Satellite data from two different platforms and ground-based monitoring data were employed to assess selected air pollutants. Additionally, various datasets were used to derive urbanization parameters such as population and land use.

Air quality data -Satellite
Modern-Era Retrospective Analysis for Research and Applications, Version 2(MERRA-2)

MERRA-2 is the newest reanalysis application of NASA for satellite-derived data from 1980 onwards, using the Goddard Earth Observing System, version 5 (GEOS-5), an Earth system model that provides data at high temporal (hourly to daily) and spatial resolutions (~50 km globally). By assimilating satellite observations, it provides an improved representation of atmospheric composition, including aerosols and air pollutants. This is useful for analyzing long-term trends. The data on atmospheric aerosols of black carbon, organic carbon, sulfate, sea salt, and dust were obtained using MERRA-2.

The $PM_{2.5}$ concentration is calculated according to the following equation 1:

$$\mathbf{PM_{2.5} = BC + OC + (SO_4 * 132.14/96.06) + dust + SS} \qquad \mathbf{(1)}$$

where BC, OC, SO_4, dust, and SS are the concentrations for individual components obtained from MERRA-2 (Buchard et al., 2017). The sulfate concentration has been recalculated assuming that sulfate is in the form of ammonium sulfate $(NH_4)_2SO_4$ (GMAO, n.d.).

Deseasonalising Method

The deseasonalising method helps remove seasonal variations from the long-term trend. By eliminating seasonal patterns, the method allows for more accurate focus on the long-term trend (Carslaw et al., 2020).

In this study, deseasonalised graphs were created using the smoothTrend() function from the openair package in R (Carslaw et al., 2020). The additive method was used, which is typically represented as equation 2 (Cleveland et al., 1990).

$$\mathbf{Y_v = T_v + S_v + R_v} \qquad \mathbf{(2)}$$

Where:

Y_v is the data
T_v is the trend component
S_v is the seasonal component
R_v is the remainder component
For v = 1 to N

To decompose the time series and extract these components, the stl() function in R was used (Carslaw et al., 2020; Kalinda, 2021). Prior to decomposition, any missing data in the series is first imputed using Kalman filter and Kalman smoothing technique (Carslaw et al., 2020).

Washington University Data

Washington University provides high-resolution global and regional $PM_{2.5}$ concentrations derived from a combination of satellite observations. It offers High Spatial Resolution Data. $PM_{2.5}$ data for Sri Lanka from 1998 to 2022 is available.

Air quality data – ground -based data
U.S. Embassy Air Quality Monitoring Data

The U.S. Embassy Air Quality Monitoring provides accurate, real-time air quality data from U.S. embassies and consulates worldwide. Embassy-based monitors measured airborne fine particulate matter ($PM_{2.5}$). Real-time air quality data collected from monitoring stations in Colombo provides daily variations in pollution levels, and this monitoring data was used to validate the above-mentioned starlight data.

Urban Development Data

Urban development indicators such as population distribution, urban and rural, sources used for cooking and energy use for lighting are gathered from the population data collected from the Department of Census and Statistics, Sri Lanka. This department provides accurate, timely statistical information on the socio-economic development of the country.

4.Results

The gathered information, including $PM_{2.5}$, Black Carbon (BC), Organic Carbon (OC), Sulphate (SO_4), dust, and sea salt (SS), was analyzed following a four-step procedure. First, the long-term annual trends of all the pollutants were compared for the period covering 1980 to 2024. Then, the variations of the pollutants were studied to explore their seasonal behavior in Sri Lanka. Following that, an analysis was conducted to examine the variations over monthly, weekly, daily, and hourly time scales with a view to understanding periodic trends along with short-term changes. Finally, a spatial comparison was conducted to examine the differences in the concentrations of the pollutants between the four chosen study sites: Colombo, Kandy, Anuradhapura, and Jaffna.

The Long-term Annual Trends Analysis

Figure 4 - 7 shows the monthly mean deseasonalised concentrations ($\mu g\ m^{-3}$) of $PM_{2.5}$ and its major components, BC, OC, SO_4, dust, and SS, in Colombo, Kandy, Anuradhapura, and Jaffna for the years 1980 - 2024 from MERRA-2 satellite dataset. The x and y axes of all plots are year (1980–2024) and concentration (in $\mu g\ m^{-3}$), respectively. Monthly data points are shown by red dots; the overall trend over time is shown by a solid line.

The presentation of deseasonalized data allows clear demonstration of the long-term trend of these pollutants, indicating their variation without any reference to normal seasonality changes. The approach employed allows precise determination of the influence of factors such as urbanization and population growth on atmospheric quality over the specified duration.

Colombo

The deseasonalised monthly mean $PM_{2.5}$ concentration for Colombo over the four decades (Figure 4A) is stable. During the time series analysis, the majority of the monthly readings are between 3 and 5 µg m^{-3}. As can be seen from Figure 4B, Black Carbon concentrations show small variations around a low background level, typically between 0.01 and 0.05 µg m^{-3}. Relative to the baseline trend, there is a moderate decrease in BC concentrations around 1991, after which the concentrations remain steady. From around 2007, BC concentrations appear to exhibit an increasing trend. The OC concentrations at Colombo (Figure 4C) record a rather increasing trend with years, especially after 2009, and concentrations are typically in the range of 0.1 to 0.3 µg m^{-3}. Sulfate concentrations at Colombo (Figure 4D) were comparatively high at the beginning of the time series. The SO_4 concentration graph presents a marked and steady decrease from 1980 to 2001, after which a period of stable low concentration persists up to 2024. During this time interval, values generally oscillate in the 0.5 - 1.0 µg m^{-3} range. Dust concentrations (Figure 4E) present moderate fluctuations, frequently between 0.2 and 0.8 µg m^{-3} across the large temporal extent. Sea Salt concentrations (Figure 4F) are flat throughout the period. The monthly readings are between 2 and 4 µg m^{-3}.

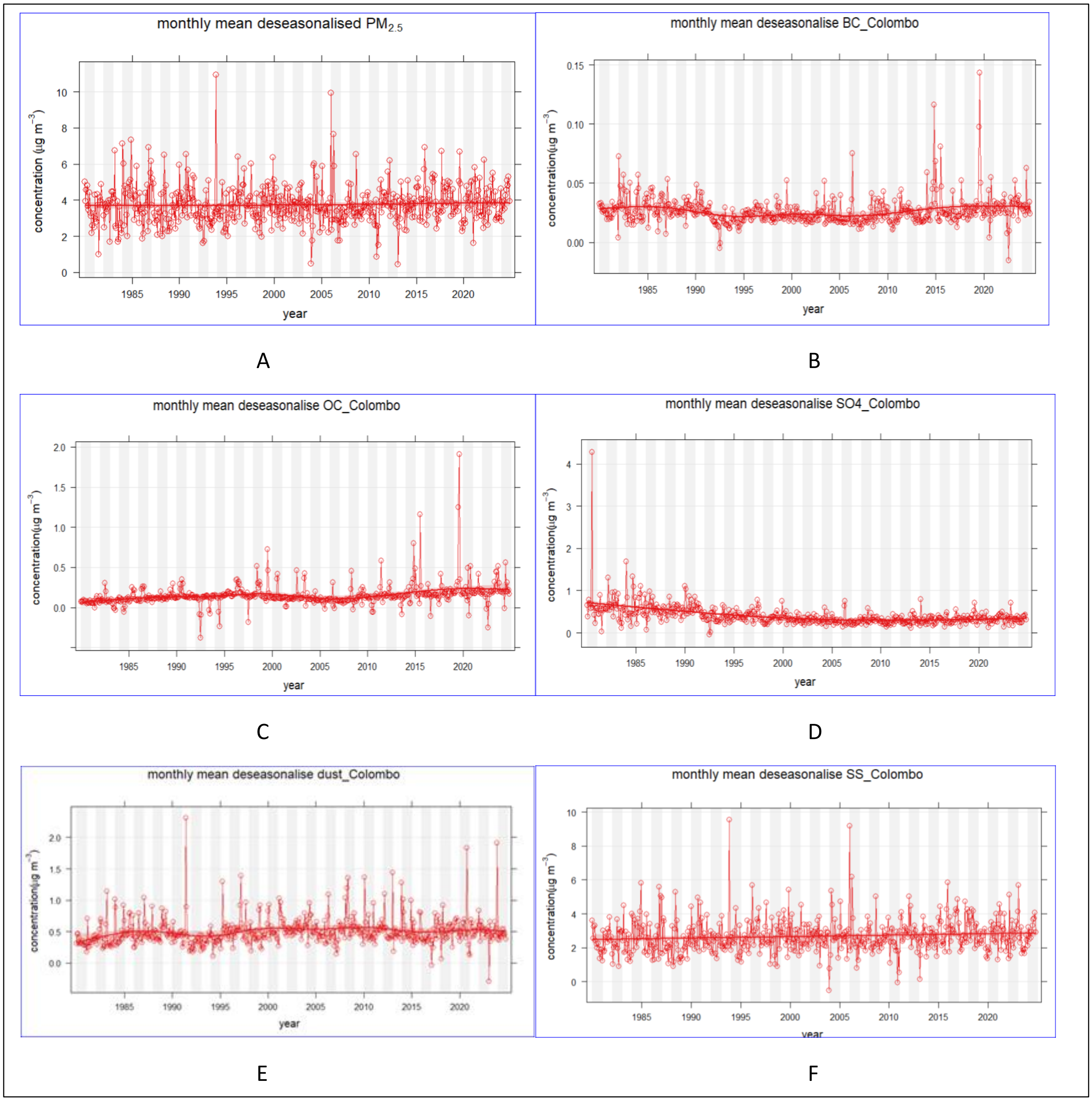

Figure 4: Monthly mean deseasonalised $PM_{2.5}$ BC, OC, SO_4, dust, and SS in Colombo from 1980 to 2024 based on MERRA-2 satellite data.

Kandy

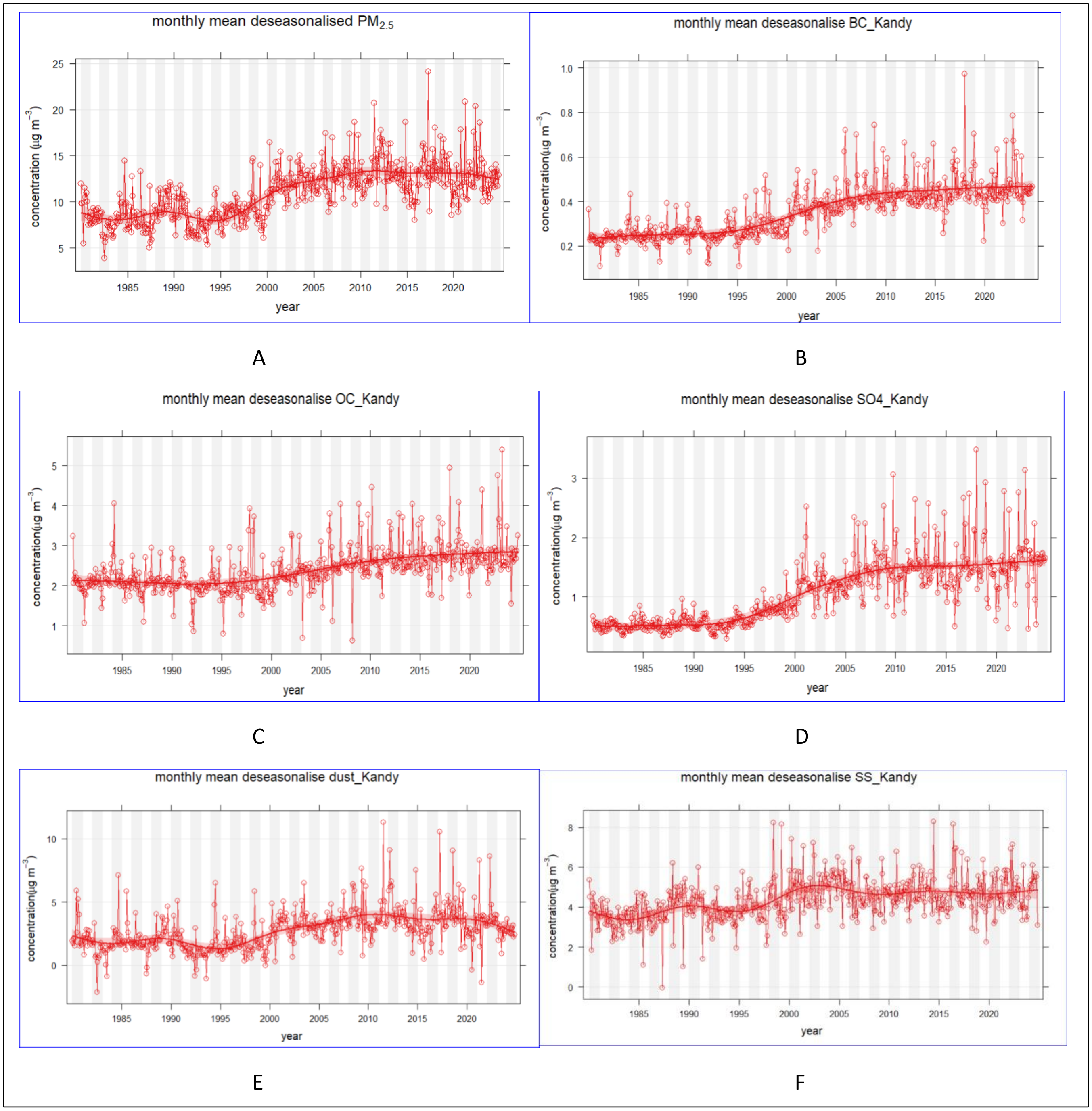

Figure 5: Monthly mean deseasonalised $PM_{2.5}$ BC, Oc, SO_4, dust and SS in Kandy from 1980 to 2024 based on MERRA-2 satellite data.

Figure 5A presents relatively stable low deseasonalized monthly mean $PM_{2.5}$ concentrations from 1980 to 1995, fluctuating from 7 to 10 µg m^{-3}. From 1995 to 2012, $PM_{2.5}$ concentrations increased from 7 to 12 µg m^{-3}. From 2015 onwards, the concentration appeared to stabilize or decrease slightly, but still fluctuated within a broader band compared to the initial decades.

As evident from Figure 5B, the deseasonalised monthly mean BC concentration in Kandy shows a detectable increasing trend during the study period of 1980-2024. The black carbon concentration exhibits a very slight increasing trend from 1980 to 1995. The measured BC concentrations were very low during this pre-1995 period and usually varied in the range of 0.2-0.3 μg m^{-3}. After 1995, there is a notable rise, with ongoing increases. The deseasonalized organic carbon (OC) concentration time series (Figure 5C) is comparatively flat over the years 1980 to 1995 within a narrow range of approximately 1.8–2.1 μg m^{-3}. After 2000, a moderate increasing trend is seen, with the smoothed line also showing a gradual increase in concentrations.

Figure 5D indicates that during a span of four decades, the concentration of SO_4 exhibits increased considerably. From 1980 to 1993, SO_4 concentration remains quite stable, with a low concentration of approximately 0.5 μg m^{-3}. Following 1993, the concentrations of sulfate exhibit a steady increasing trend. Dust concentrations (Figure 5E) exhibit a generally flat pattern from 1980 to 2000, fluctuating between 2 and 4 μg m^{-3}. Beginning in 1999, the monthly average concentration of dust started a consistent upward trend. From 2005 to 2015, levels reached historic highs. Following 2015, the trend would seem to indicate a modest decrease. According to Figure 5F, from 1980 to 1997, SS concentrations were fairly stable, generally fluctuating between 3.5 and 4.5 μg m^{-3}. In the late 1990s to early 2000s, a steady increase began, with the smoothed line showing that the baseline concentrations rose. Concentrations of sea salt from 2000 to 2024 are stable.

Anuradhapura

The deseasonalised monthly mean $PM_{2.5}$ concentration (Figure 6A) shows relatively consistent variations over a span of four decades. Over this time span, the mean $PM_{2.5}$ concentration generally varies between 2 and 5 μg m^{-3}. Black carbon concentrations (Figure 6B) continue to maintain low values throughout the observation period, with most of the monthly values falling below 0.05 μg m^{-3}. Organic carbon concentrations (Figure 6C) show a regular long-term pattern, with monthly means normally ranging from 0.2 to 0.6 μg m^{-3}. There is a slight increase after 2000; however, this change is not extreme. Sulfate concentrations (Figure 6D) have a clear decreasing trend. The early part of the record (1980s) shows higher concentrations. Deseasonalized mineral dust concentrations (Figure 6E) are relatively constant throughout the period. The trend line has little variability with no significant increase or decline. Sea salt aerosol concentrations (Figure 6F) have a stable trend over the study period. Monthly values range from 1 to 4 μg m^{-3}.

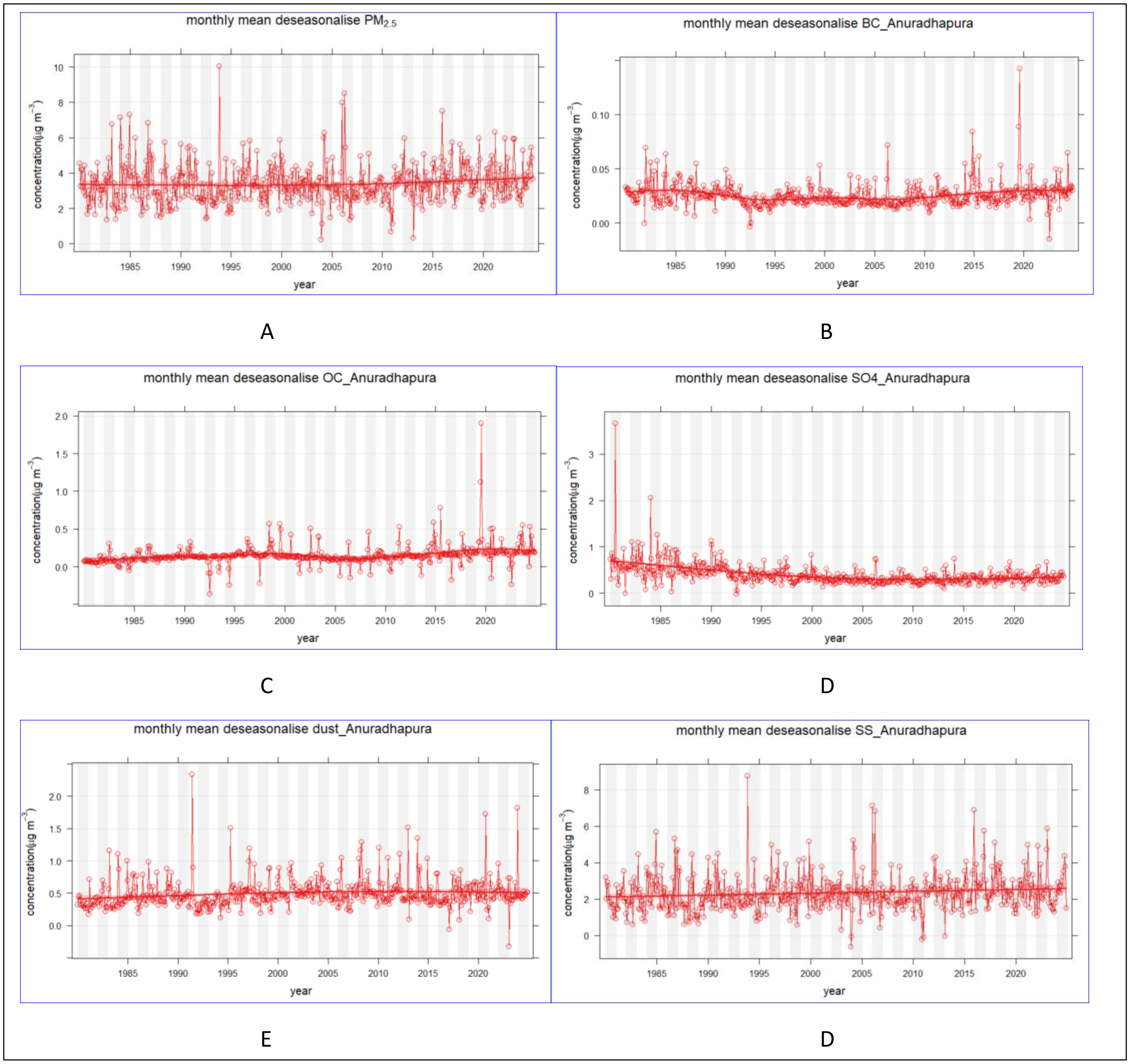

Figure 6: Monthly mean deseasonalised $PM_{2.5}$, BC, OC, SO_4, dust, and SS in Annuradhapura from 1980 to 2024 based on MERRA-2 satellite data

Jaffna

As evident in Figure 7A, the deseasonalized $PM_{2.5}$ concentrations in Jaffna trace a generally stable long-term trajectory with modest variability ranging between 2 and 5 µg m^{-3}. On the whole, the smoothed trend indicates that $PM_{2.5}$ concentrations have remained relatively stable over the decades. From Figure 7B, Black carbon levels in Jaffna have been low throughout the observation period, generally below 0.05 µg m^{-3}. There is a weak declining trend from the early 1980s to the mid-2000s, after which a weak increase or leveling off is noted in recent years. The deseasonalized organic carbon (OC) levels (Figure 7C) show a slowly rising trend with time. Beginning at concentrations of approximately 0.3 to 0.4 µg m^{-3} in the early decades, the smoothed trend rises to concentrations of approximately 0.5 to 0.6 µg m^{-3} in the most recent decades. Sulfate levels (Figure 7D) have experienced a long-term decrease throughout the period of study. Deseasonalised dust levels (Figure 7E) show a fairly consistent pattern across the years, with monthly means generally ranging from 0.5 to 1.5 µg m^{-3}. Sea salt aerosol levels.

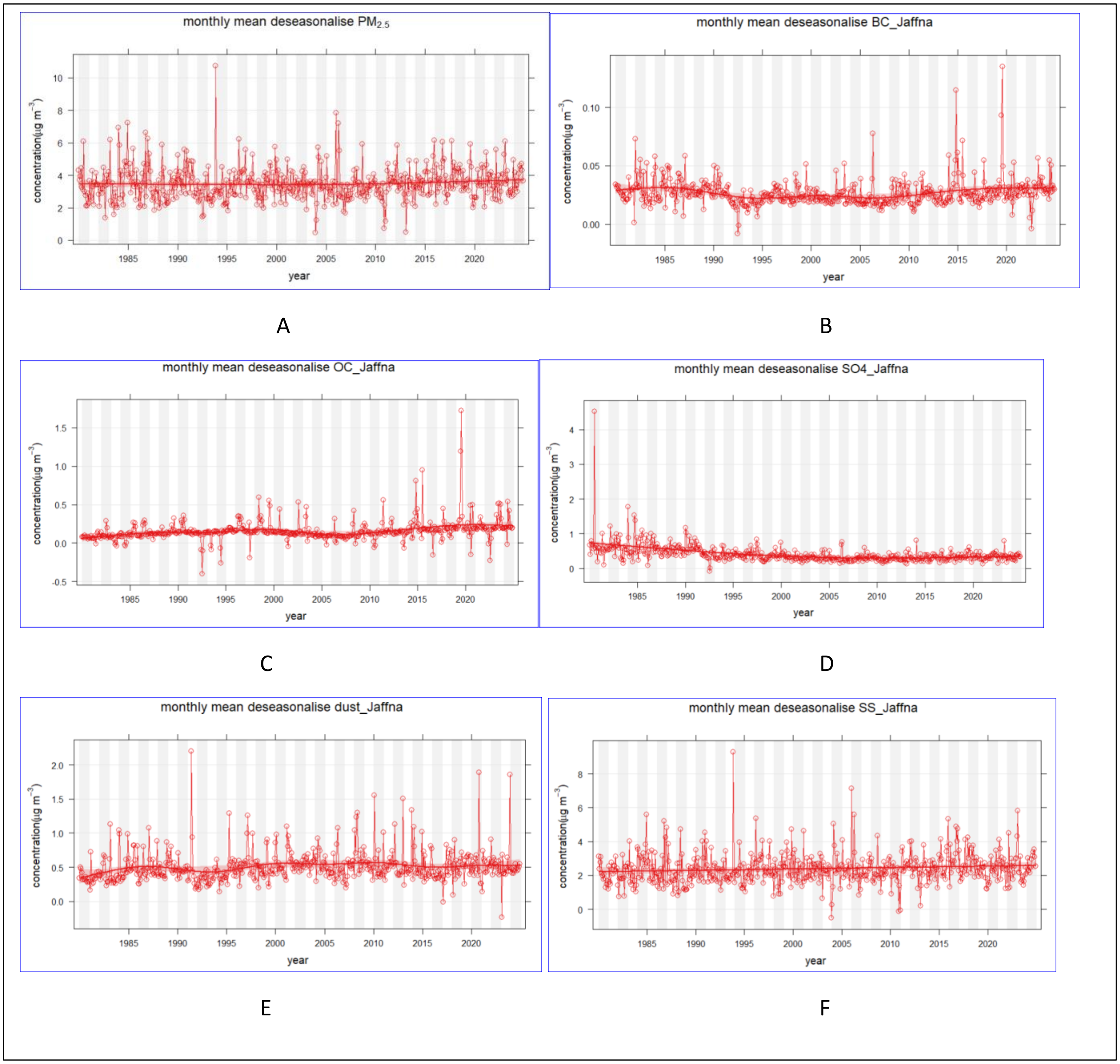

Figure 7: Monthly mean deseasonalised $PM_{2.5}$, BC, OC, SO_4, dust, and SS in Jaffna from 1980 to 2024 based on MERRA-2 satellite data

Normalizing Time Series

Figures 8 - 11 present the normalized annual trends of $PM_{2.5}$ and its constituents including BC, OC, SO_4, Dust, and SS from 1980 to 2024. By utilizing a normalized time series, it is meant that data for all the pollutants have been converted to a common scale, and hence, an easy comparison between their relative changes with time independent of differences in their original units or magnitude is possible. This method effectively illustrates the variation of each pollutant relative to the observed value in 1980, rather than its absolute levels. This relative method makes long-term trends and variability among pollutants easier to comprehend when plotted on the same graph.

Colombo

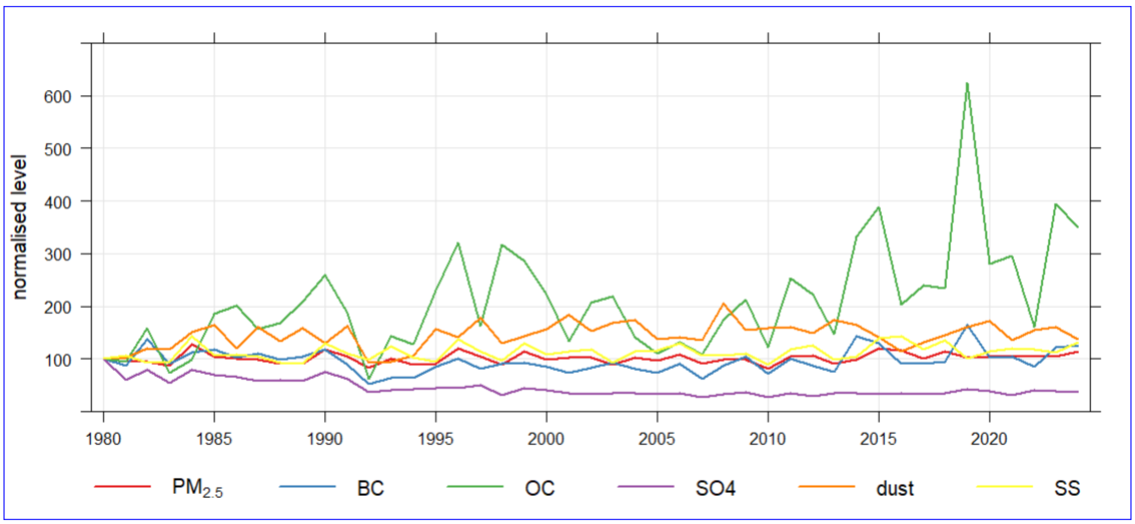

Figure 8: Normalizing time series data in Colombo to fix values to equal 100 at the beginning of 1980

According to Figure 8, $PM_{2.5}$, BC, dust, and SS are steady throughout the period. Organic Carbon increases during a long-term trend. The significant rise in biomass burning, trash burning, or seasonal emissions increases OC. Sulfate decreases throughout the period, suggesting either effective control of sulfur emissions or minimal change in sulfate aerosol formation.

Kandy

According to Figure 9A, SO_4 shows the most rapid and sustained increase, especially after 2000, rising to more than three times its earlier levels. This implies an increasing influence of secondary sulfate aerosol formation or regional transport mechanisms. Black Carbon shows steady increasing trends, implying contributions from urban emissions, combustion sources, and traffic pollution. $PM_{2.5}$ in Kandy slightly increased during this study period. Figure 9B strongly suggests that SO_4 and BC are the prime drivers of overall $PM_{2.5}$ enhancement over the decades in Kandy. It also shows that $PM_{2.5}$, BC, and SO_4 all increase consistently over time. This further highlights the growing dominance of secondary pollutants (SO_4) and combustion sources (BC) in recent decades.

Figure 9A and 9C show that OC and dust exhibit greater year-to-year variability but are more constant in the long term compared to SO_4 and BC. Dust is episodic with high interannual variability. But before 2000, dust concentration was steady and low compared to after 2000, as might be expected for resuspended soil or construction particles. Additionally, it could be caused by imperfections in the reanalysis data. Specifically, data from after 2000 are more reliable due to the introduction of the MODIS instrument on satellites. On the other hand, OC and SS present a steady but slowly increasing trend, with OC displaying a good long-term increase that could be associated with urban biomass or organic emissions. Whereas SS presents a mostly stable trend, indicating a sustained contribution from marine sources.

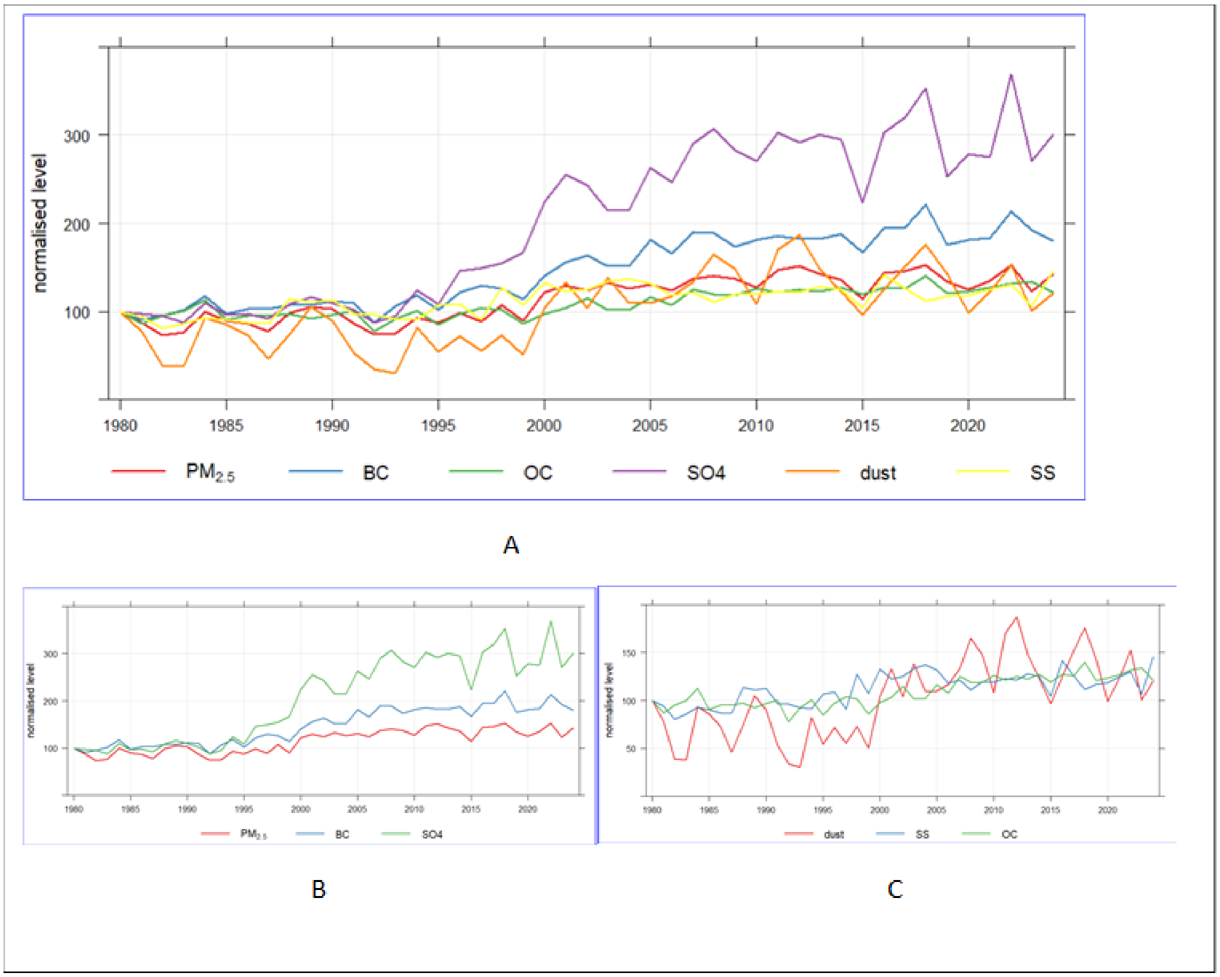

Figure 9: :(A) Normalising time series data in Kandy to fix values to equal 100 at the beginning of 1980 for $PM_{2.5}$., BC, OC, SO_4, dust, and SS (B) Normalising time series data in Kandy to fix values to equal 100 at the beginning of 1980 for $PM_{2.5}$., BC, and SO_4 (C) Normalising time series data in Kandy to fix values to equal 100 at the beginning of 1980 for dust., SS,, and OC

Anuradhapura

The overall $PM_{2.5}$ line shows a stable trend. OC has the most dynamic and apparent changes over time. It is modestly high throughout the 1980s and 1990s, and then is more liable to show more frequent and higher-amplitude fluctuations after 2000, with spectacular peaks in particular near 2015 and 2020, where it surpasses a 600% rise from baseline in certain years. Episodic emissions, such as from biomass burning, waste burning, or heightened vehicular and industrial activity, are strongly implied by this trend. Dust concentrations are relatively stable before 2000, with a slight increasing trend afterward. Peaks are small but consistent, indicating chronic resuspended dust sources such as construction, unpaved roads, and land degradation. Black Carbon is generally stable throughout the study period. Stability would suggest steady combustion-related emissions, e.g., diesel engines, domestic fuels, with no radical change over the decades, but a modest long-term increase indicating growing urban activity. SO_4 shows a decreasing trend, with a nearly 50% reduction from 1980 to 2024. Sea salt is marked by low variability, close to the baseline, suggesting that it is more influenced by long-term coastal meteorology than by anthropogenic emissions.

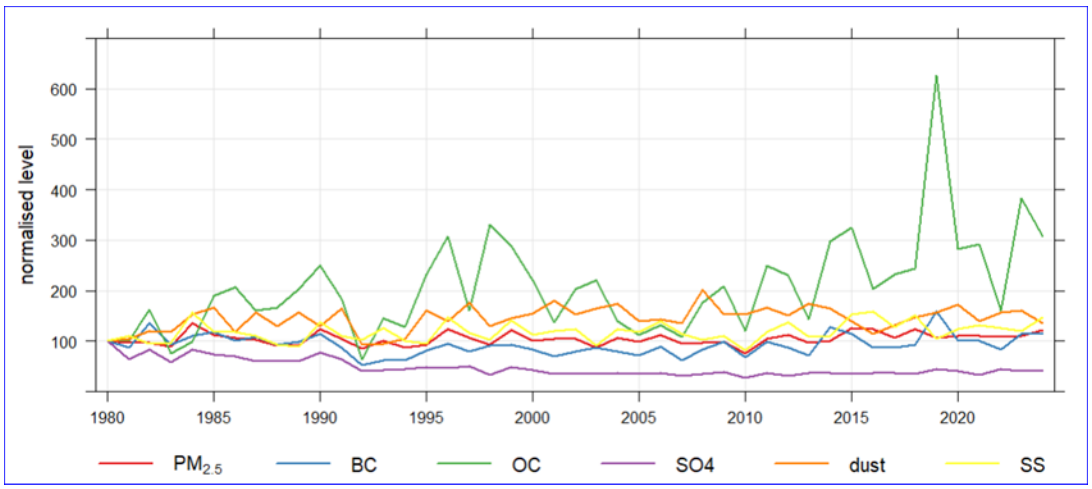

Figure 10: Normalising time series data in Anuradhapura to fix values to equal 100 at the beginning of 1980.

Jaffna

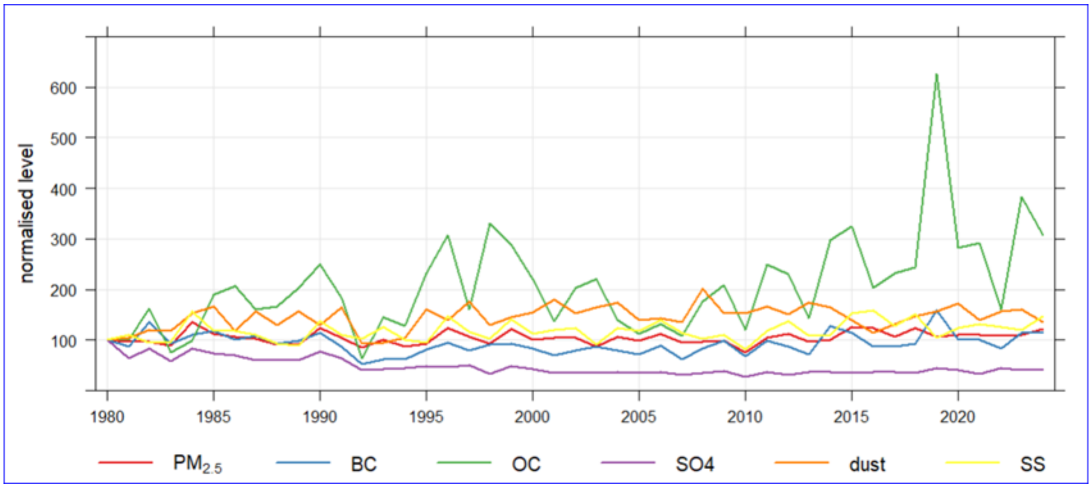

Figure 11: Standardized time series data in Jaffna to obtain values corresponding to 100 at the start of 1980.

From Figure 11, $PM_{2.5}$ concentration has relatively constant values from 1980 to 2024. Organic Carbon shows the greatest fluctuations and changes throughout the period. This indicates that OC is the most highly variable contributor to $PM_{2.5}$ in the past years in Jaffna. Dust is moderately variable and consistently at higher levels than most of the other components, both before and after 2000. It is in a relatively stable but high pattern, suggesting its steady contribution. Black Carbon shows a steady trend throughout the period examined, with a slight upward trend visible after 2000. The trend points to a steady increase in emissions from combustion sources, particularly from vehicles. SO_4 shows a decreased trend across the period viewed, with a lower value compared to 1980; this could be due to regulated or decreasing sulfur emissions or intrinsic stability in its source origins. Sea Salt levels are low and constant, with small variability, indicative of meteorological or seasonal influences, rather than anthropogenic peaks.

Annual Mean $PM_{2.5}$ trend Analysis derived from Washington University's satellite data from 1998 through 2022

Figure 12 - 15 shows the yearly average $PM_{2.5}$ concentrations between the years 1998 and 2022 based on Washington University satellite-derived data. The color scale is from low values of violet to high values of orange, and the values extend from 0 to 40 $\mu g\ m^{-3}$ in Colombo, Kandy, Anuradhapura, and Jaffna. Each map depicts a single year, enabling the visualization of long-term temporal changes in fine particulate matter.

Colombo

The time series of spatial concentration maps (Figure 12) illustrates a noticeable trend in $PM_{2.5}$ concentrations across Colombo from 1998 to 2022. In the late 1990s and early 2000s, the levels of $PM_{2.5}$ were relatively low. Over time, however, the concentration of $PM_{2.5}$ pollution shows relatively increased. The reduction or stabilization appears in 2021-2022. Urban expansion, and increased anthropogenic emission appear to be drives of air quality in Colombo over time.

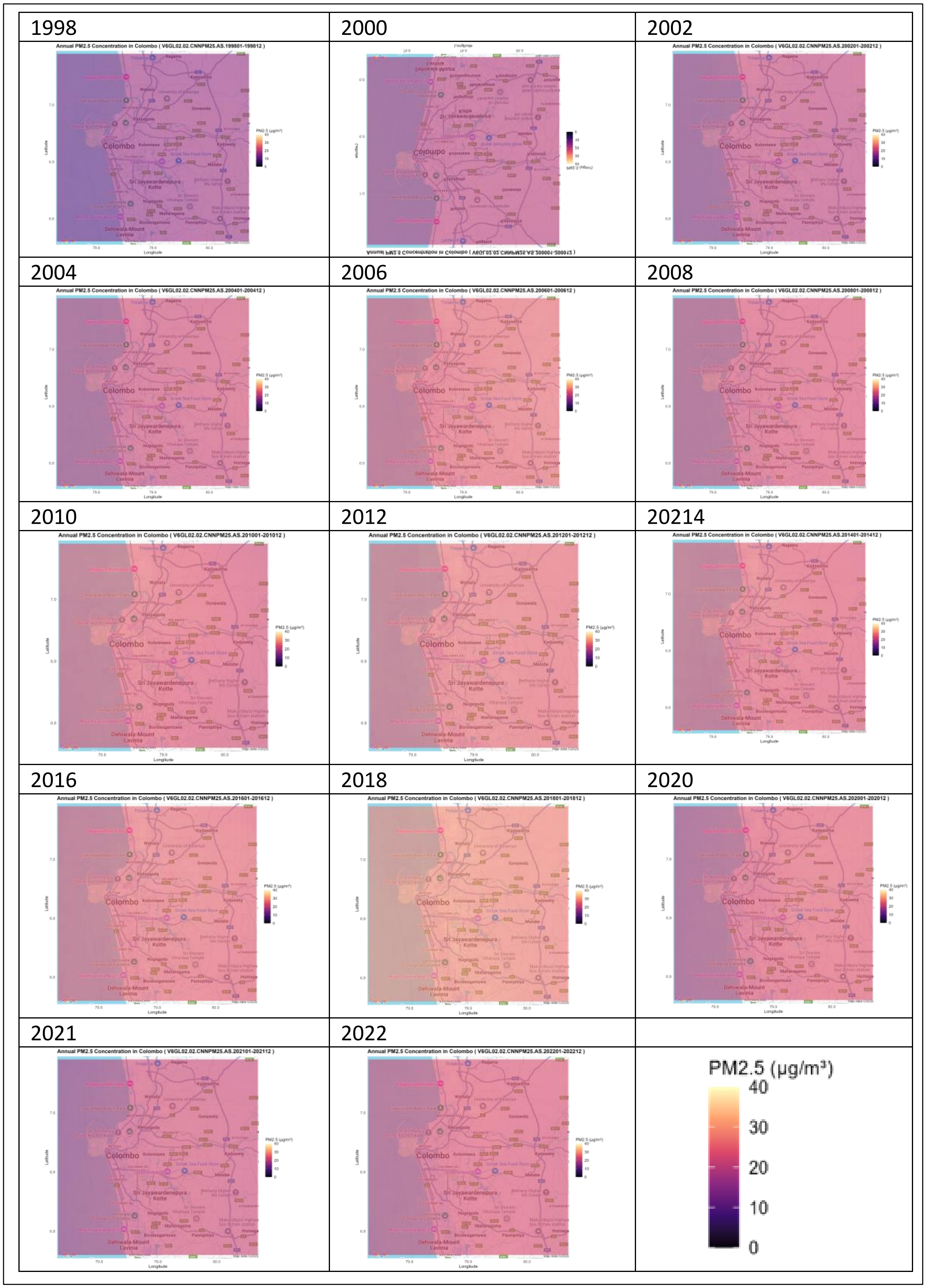

Figure 12: Annual mean $PM_{2.5}$ concentration in Colombo from 1998 to 2022

Kandy

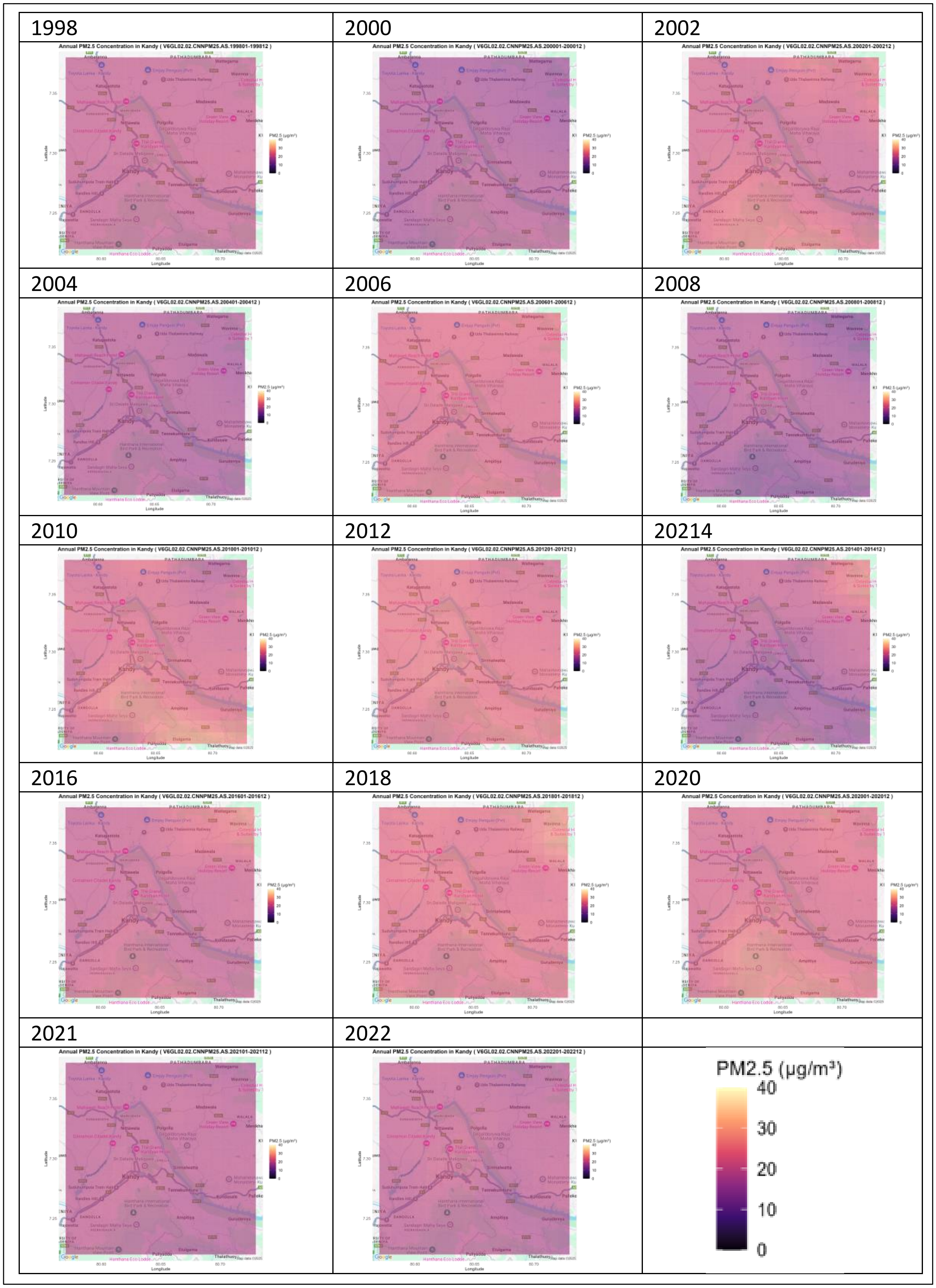

Figure 13: Annual mean $PM_{2.5}$ concentration in Kandy from 1998 to 2022.

According to Figure 13, the trend analysis of $PM_{2.5}$ level in Kandy from 1998–2022 indicates noticeable changers over time. The higher concentration of $PM_{2.5}$ are observed around the city center throughout the study period, possibly due to traffic emission and other anthropogenic emission. Additionally, relatively high levels also appear around the mountain region of Hanthana, which may be pollutant trapping due to the topography. Between 2004 and 20201 there is a noticeable intensification in $PM_{2.5}$. This is possibly due to urbanization and rising vehicular traffic in the city.

Anuradhapura

Figure 14 shows the increasing trend of $PM_{2.5}$ concentration in Anuradhapura during the 1998-2022 period. At the beginning, in the late 1990s, the concentration of $PM_{2.5}$ in the region was comparatively low, reflecting comparatively clean air. However, from the mid-2000s, $PM_{2.5}$ level increases. From the year 2006, the concentration increased in magnitude and expanded in area, covering larger and larger parts of the city on a continuous basis. The last two years 2021 and 2022, the concentration shows reduction of emissions.

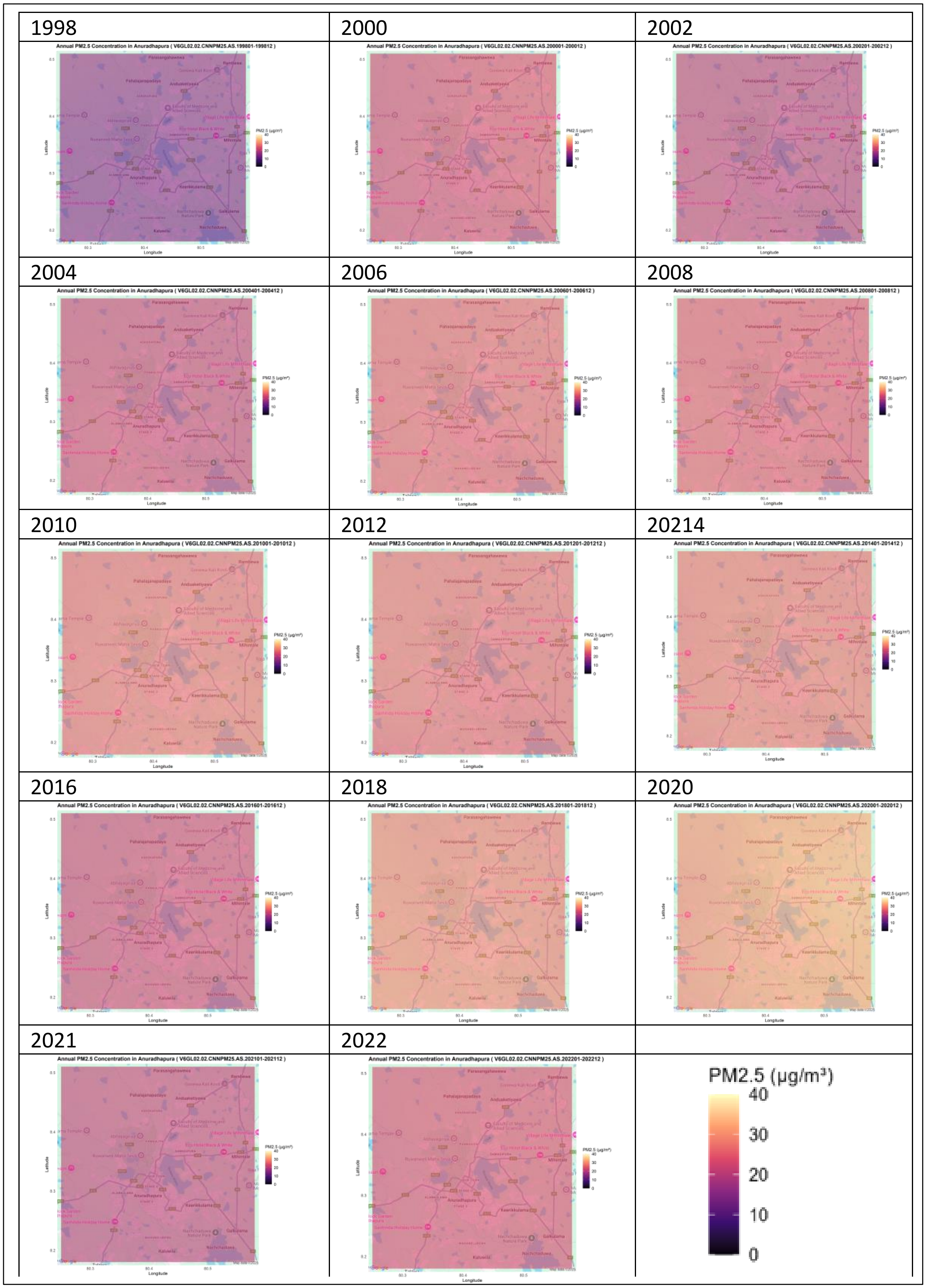

Figure 14: Annual mean $PM_{2.5}$ concentration in Anuradhapura from 1998 to 2022.

Jaffna

According to Figure 15, $PM_{2.5}$ the level increased from 1998 to 2010. In the late 1990s, the $PM_{2.5}$ concentrations were relatively low. From 2004, there is a gradual increase of $PM_{2.5}$. The magnitude and spatial distribution of PM level is significantly higher across the region in 2010. This sharp increase may be due to post-war reconstruction, an increase in population returning or migrating to Jaffna city and increase vehicle movement after end of the civil war in 2009.

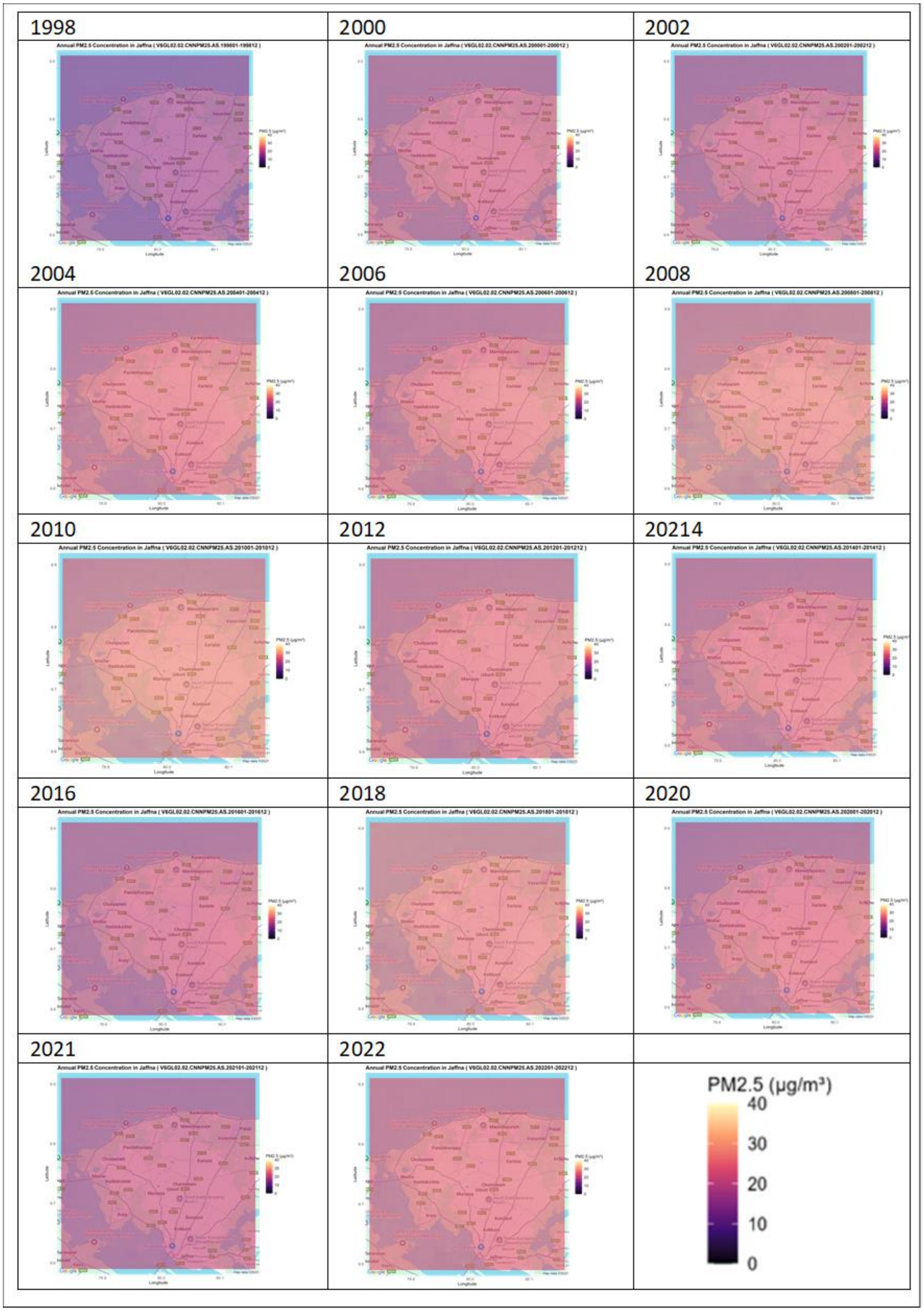

Figure 15: Annual mean $PM_{2.5}$ concentration in Jaffna from 1998 to 2022

Validation of $PM_{2.5}$ Data Sources Using Observations from the U.S. Embassy in Colombo

In the study, satellite data of MERRA-2 (1980-2024) is employed primarily to gather long-term atmospheric data on $PM_{2.5}$, BC, OC, SO_4, dust, and SS. To enhance the precision of $PM_{2.5}$, Washington University satellite data (1998-2022) is also used. In addition to this, ground-level measurements of $PM_{2.5}$ at the US Embassy in Colombo (2017-2024) are also employed to cross-validate satellite-derived data to make the study more robust and reduce the error linked to satellite data.

The notable differences in absolute concentration value are observed by comparing $PM_{2.5}$ concentrations from Satellite -derived datasets and ground base monitoring data. This is due to the fundamental differences in measurement. MERRA-2 and Washington University satellite data provide spatial averages over broader grid cells. The US Embassy in Colombo (ground-based data) provides point specific measurements. Therefore, satellite data reports lower concentration compared to ground-based data.

Figure 16B clearly illustrates a declining trend in $PM_{2.5}$ concentration in Colombo between 2018 and 2021 and between 2022 and 2024, while showing an increasing pattern from 2021 to 2022. Figure 16A and Figure 16 B depict a decrease of $PM_{2.5}$ level from 2018 to 2019, while Figure 16C illustrates an increasing pattern. From 2019 to 2020, Figure 16A shows inclined but Figure 16 B and 16C shows a decline of PM level. Figures 16A, 16 B, and 16C, depict $PM_{2.5}$ concentration drop from 2020 to 2021. Both Figure 16B and 16C, illustrate the $PM_{2.5}$ concentration increase from 2021 to 2022 while Figure 16A shows a steady pattern. These shared patterns suggest that while numerical values differ, the directional trends are consistent. This cross-validation increases the suitability of using MERRA-2 and Washington University datasets for long-term air quality assessments.

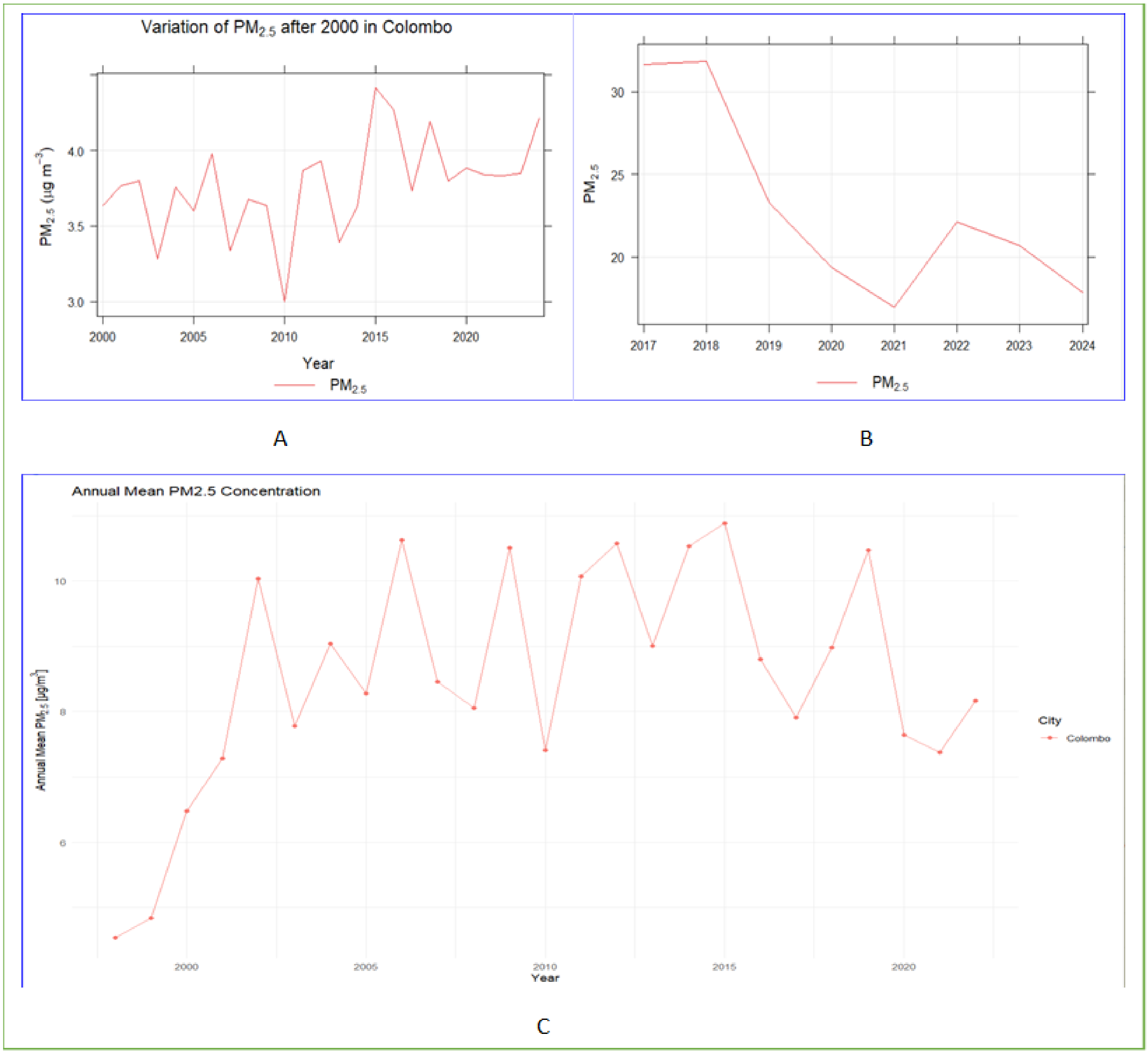

Figure 16: A: variation of PM2.5 in Colombo from 2000 to 2024 based on MERRA-2, B: variation of PM2.5 from 2017 to 2024 in Colombo based on the US Embassy in Colombo, and C: variation of Annual mean PM2.5 concentration in Colombo using Washington University

Variation of $PM_{2.5}$, BC, OC, SO_4, dust and SS with monsoon seasons in Sri Lanka

Sri Lanka experiences two main monsoons: the southwest monsoon and the northeast monsoon, along with two intermonsoons. In this section, the analysis of the behavior of $PM_{2.5}$ and its components with respect to the effects of monsoon seasons is presented.

Colombo

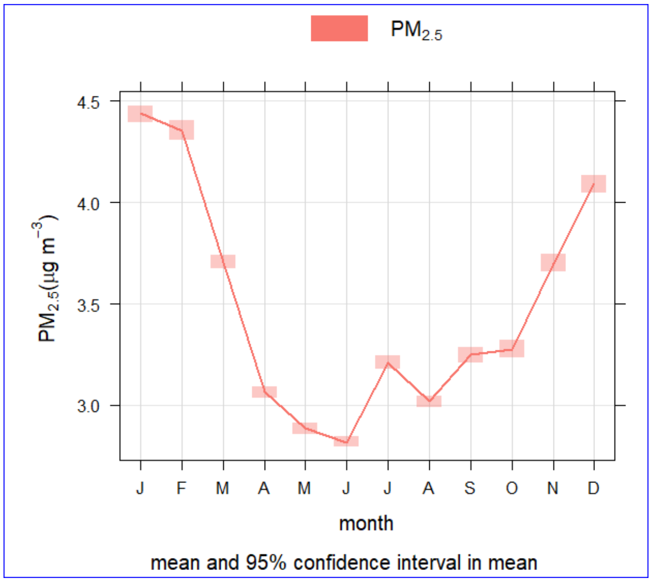

Figure 17: $PM_{2.5}$ monthly variation in Colombo

The $PM_{2.5}$ variations throughout the year in Colombo, Sri Lanka, exhibit a definite annual trend based on the local monsoon-influenced climate, as indicated by Figure 17. During the northeast monsoon period and especially during January and February, $PM_{2.5}$ concentrations are at their highest, reaching nearly 4.5 µg m^{-3}. This season is generally marked by dry, stable atmospheric conditions over Colombo, which hinder the natural dispersion of air pollutants. The low rainfall and lighter winds characteristic of this time of year permit the fine particles from car exhaust, residential combustion, and other localized sources of emission to build up in the air.

After this peak, $PM_{2.5}$ concentrations (Figure 17) show a significant decrease from March to May, in accordance with the onset of the first inter-monsoon season that introduces more erratic wind regimes and greater possibilities of convection and precipitation.

Minimum levels are found during the southwest monsoon period (May-August), by which time $PM_{2.5}$ levels drop to below 3.0 µg m^{-3}. The intense monsoonal winds together with regular rainfall during this period facilitate the dispersion and removal of pollutants, thereby enhancing the general air quality in Colombo.

Also, Figure 17 indicates that from September, $PM_{2.5}$ concentrations start to rise in the subsequent inter-monsoon (October–November) and stretch through December, which is followed by declining precipitation and increasing atmospheric stability. Concentrations recover to around 4.1 µg m^{-3} by December, which heralds the onset of another accumulation phase that is ushered in by the dryness characteristic of the northeast monsoon.

Kandy

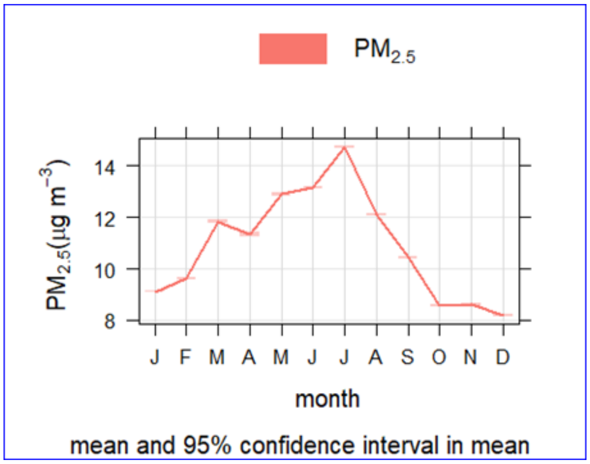

Figure 18: $PM_{2.5}$ monthly variation in Kandy

Seasonality of $PM_{2.5}$ concentrations in Kandy, Sri Lanka, is in a characteristic pattern governed by the geographical location of the region and monsoon climate. Figure 18 shows that $PM_{2.5}$ level increase from January to July, suggesting a buildup of pollutant during the Northeast monsoon period Different from the coastal regions, Kandy is an inland mountain city and it experiences its maximum $PM_{2.5}$ concentrations during the southwest monsoon season with a peak concentration of over 14 µg m^{-3} in July. This is a peculiar trend when compared with other parts of Sri Lanka, where air pollution typically drops with the rainy southwest monsoon.

Anuradhapura

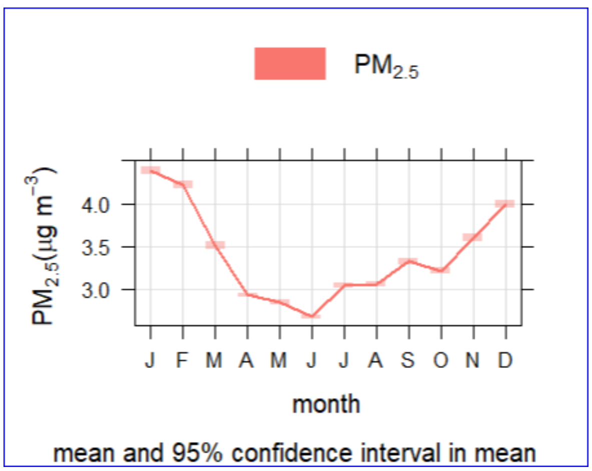

Figure 19: $PM_{2.5}$ monthly variation in Anuradhapura

As evident in Figure 19, the seasonal variation of $PM_{2.5}$ concentrations at Anuradhapura, Sri Lanka, portrays the influence of the monsoonal climate of the island. $PM_{2.5}$ concentrations are highest during the northeast monsoon period, especially December, January, and February, with the values generally exceeding 4.0 µg m^{-3}. The period concerned is characterized by dry and calm climatic conditions in the North Central Province, which limits the natural removal and dispersal of airborne particulates. In effect, this promotes the build-up of pollutants from sources like vehicle emissions, domestic combustion, and dust in the air.

With the onset of seasons into the first inter-monsoon period (March–April), the $PM_{2.5}$ concentration decreases noticeably. This decline continues with the onset of the southwest monsoon (May–September), which ensures that $PM_{2.5}$ concentrations reach their lowest point in the months of June and July. The decline is due to increased precipitation and strong southwesterly winds, which help in eradicating particulate matter from the air and aiding in atmospheric mixing, which in turn leads to better air quality within this period.

It starts in August, with $PM_{2.5}$ levels progressively increasing during the second inter-monsoon period, which lasts from October to November. Low precipitation and high wind regime variability result in reduced atmospheric capacity to disperse and flush out pollutants. By December, the onset of the northeast monsoon brings a return to high levels of $PM_{2.5}$, thereby closing the seasonal cycle.

Jaffna

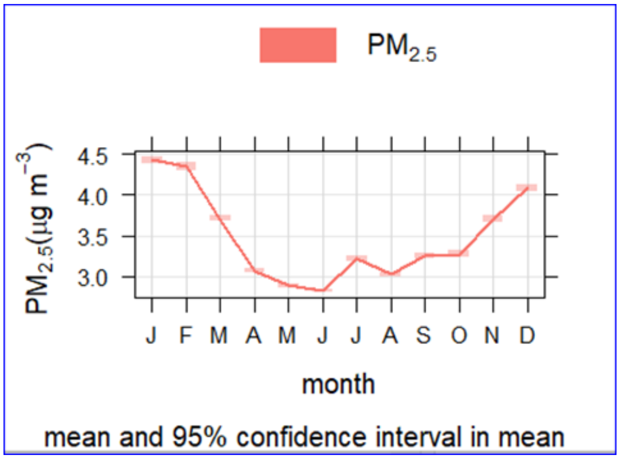

Figure 20: $PM_{2.5}$ monthly variation in Jaffna

Figure 20 shows that the monthly trend shows the highest levels of $PM_{2.5}$ are felt at the start of the year (January to March) and slowly decrease during the monsoon months. Jaffna's profile is comparatively moderate throughout the year because it is a coastal area where sea breezes allow for better dispersion of pollutants compared to inland cities.

The seasonal pattern of monthly $PM_{2.5}$ concentration variations that is apparent in Jaffna indicates a characteristic pattern governed by the monsoonal climate of Sri Lanka. Peak $PM_{2.5}$ concentrations are recorded during the northeast monsoon period, especially during January and February, with concentrations reaching around 4.5 µg m^{-3}. This is a generally dry and stable season in the northern part of the country, which may promote restricted atmospheric dispersion and buildup of pollutants in the vicinity of the surface.

As the calendar year moves to the first inter-monsoon season (March–April) and southwest monsoon months (May–September), $PM_{2.5}$ concentrations exhibit a sharp decrease. The minimum concentrations are found from May to July, when the values drop to around 3.0 µg m^{-3}. This decrease will be due to the enhanced precipitation and higher winds related to the southwest monsoon, which favor the cleaning of the atmosphere and do not allow the accumulation of fine particulate matter.

From September through the second inter-monsoon to the start of the northeast monsoon, $PM_{2.5}$ is seen to increase. Concentrations are greater than 4.0 µg m^{-3} by December, mirroring the enhancement of dry conditions less conducive for the dispersion of particulate matter. This increase also may be due to agricultural burning, household emissions, and transboundary pollution from northern India and also from the Bay of Bengal region during dry months.

In conclusion, particulate matter $PM_{2.5}$ in Jaffna has a distinct seasonal pattern with lower levels during the wet southwest monsoon and elevated levels during the dry northeast monsoon. The pattern depicts the influence of precipitation, wind direction, and local emission sources on the annual variation in air quality.

Diurnal Analysis

MERRA-2 data uses Coordinated Universal Time (UTC) for all its time stamps. So, for Sri Lanka time conversion is UTC + 5:30 hours

Colombo

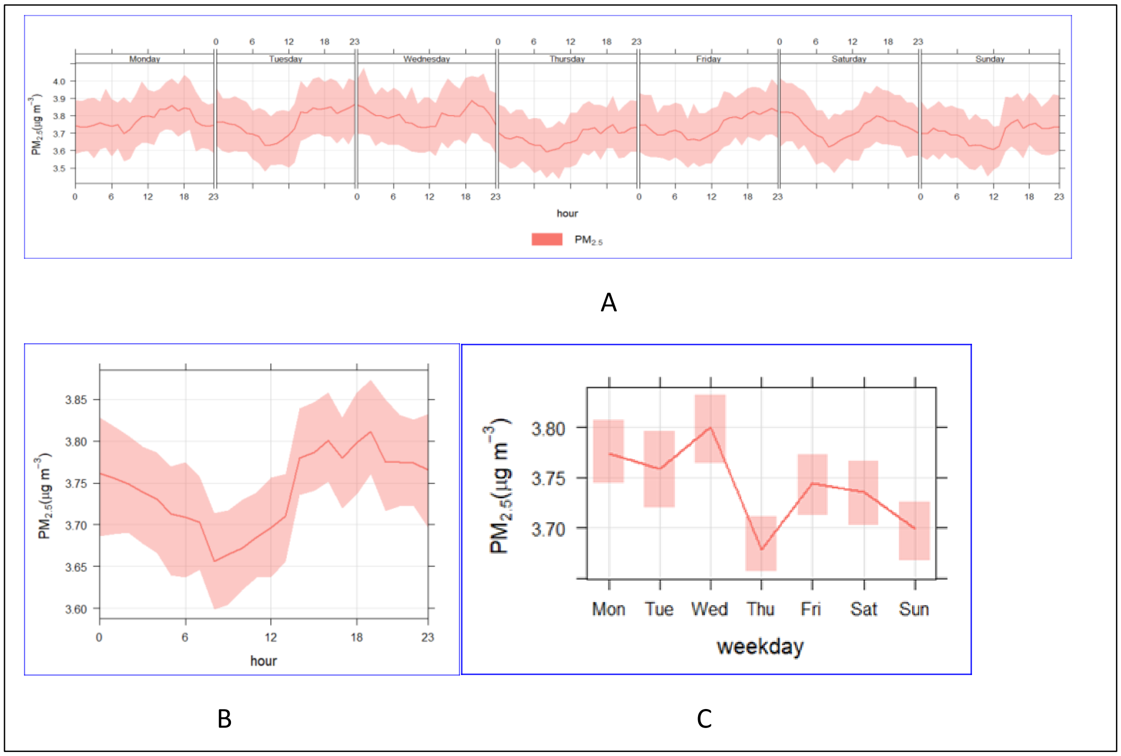

Figure 21: Diurnal variation of $PM_{2.5}$ in Colombo

From Figure 21, the hourly and daily variation of $PM_{2.5}$ concentrations in Colombo, Sri Lanka, shows how the air quality varies throughout the day and week as a result of human activity patterns and atmospheric conditions. Hourly variations for all days of the week are given in Figure 21A.

On a normal weekday, $PM_{2.5}$ concentrations in Colombo show a distinct diurnal pattern, the levels start to go up in the noon in local time (7.00+ 5:30 hours) and peak at around mid-night in local time (18+ 5:30). In the morning and early afternoon, concentrations of $PM_{2.5}$ decrease, which is probably because of enhanced atmospheric mixing and dispersion caused by rising temperatures and breezes.

There is a minor difference when the various days of the week are compared. Thursday and Sunday have relatively lower concentrations of $PM_{2.5}$, whereas Wednesday has the highest concentration throughout the week. The weekend (Saturday and Sunday) has relatively lower concentrations than weekdays, indicating a reduction in emissions from transportation and industrial activities.

Kandy

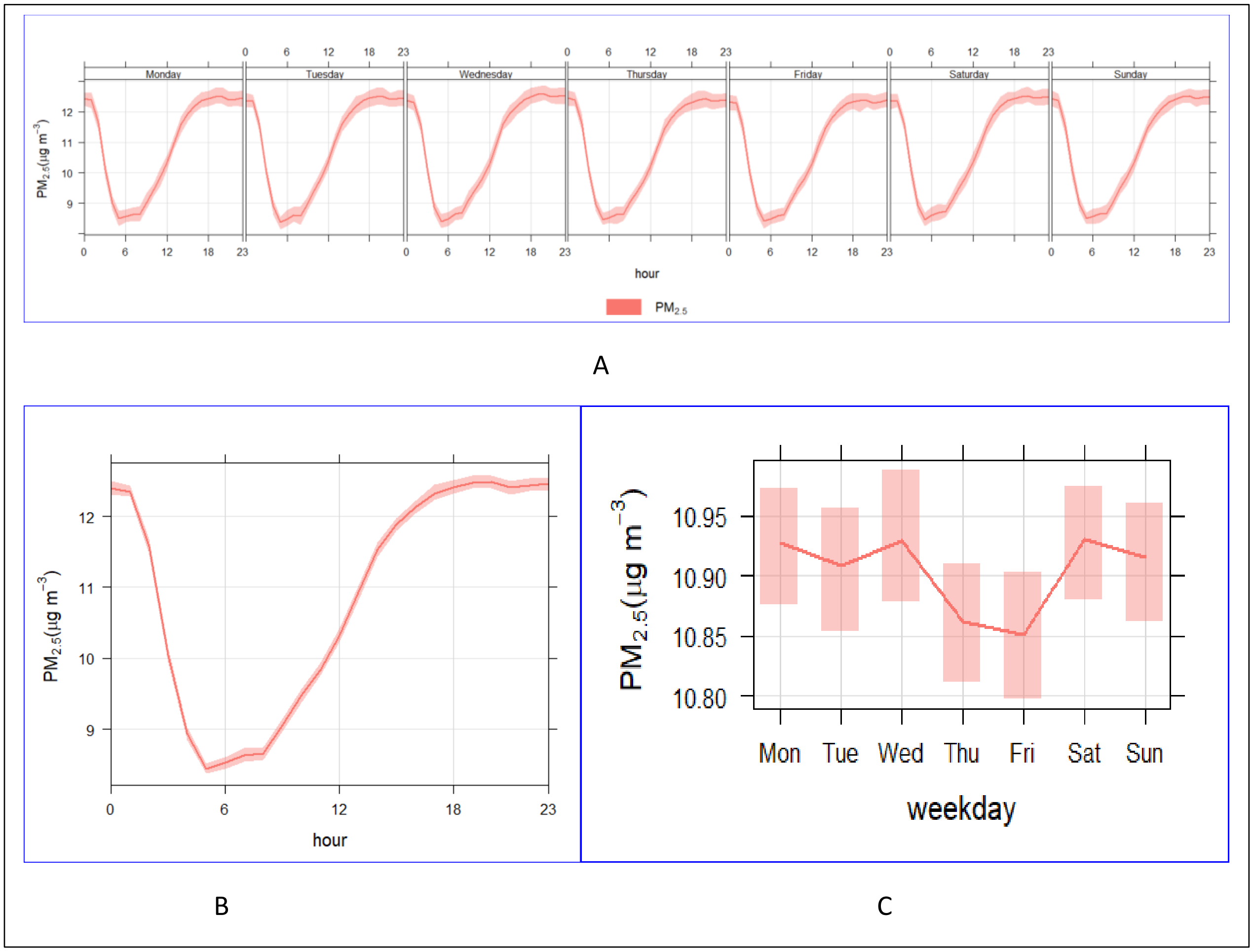

Figure 22:Diurnal variation of $PM_{2.5}$ in Kandy

The diurnal and weekly variation of the $PM_{2.5}$ levels in Kandy, Sri Lanka (Figure 22) shows a distinct pattern that is highly regulated by local topography and diurnal human activity. Figure 22A shows a typical diurnal cycle for all days of the week, while Figure 22B and 22C show total hourly and weekday averages.

On an hourly basis, $PM_{2.5}$ levels in Kandy follow a high-low-high pattern throughout the day. Concentration levels are greater predominantly at the night time and early morning hours in local time. This is probably a result of the lower temperatures at night, which minimize vertical mixing and confine pollutants near the earth's surface—a situation that is commonly enhanced in valley areas such as Kandy. $PM_{2.5}$ Concentration shows minimum during the early afternoon in local time (6+5.30).

The weekly trend indicates that $PM_{2.5}$ concentrations are fairly stable throughout the week, with slight decreases on Thursdays and Fridays, and modestly high concentrations during weekends, especially Saturdays and Sundays. This observation indicates that weekday and weekend traffic pattern differences in Kandy are less typically pronounced than in larger cities such as Colombo.

The increased weekend rates can be explained by more local trips, domestic burning, or tourist activity, which are common in and around the city.

Anuradhapura

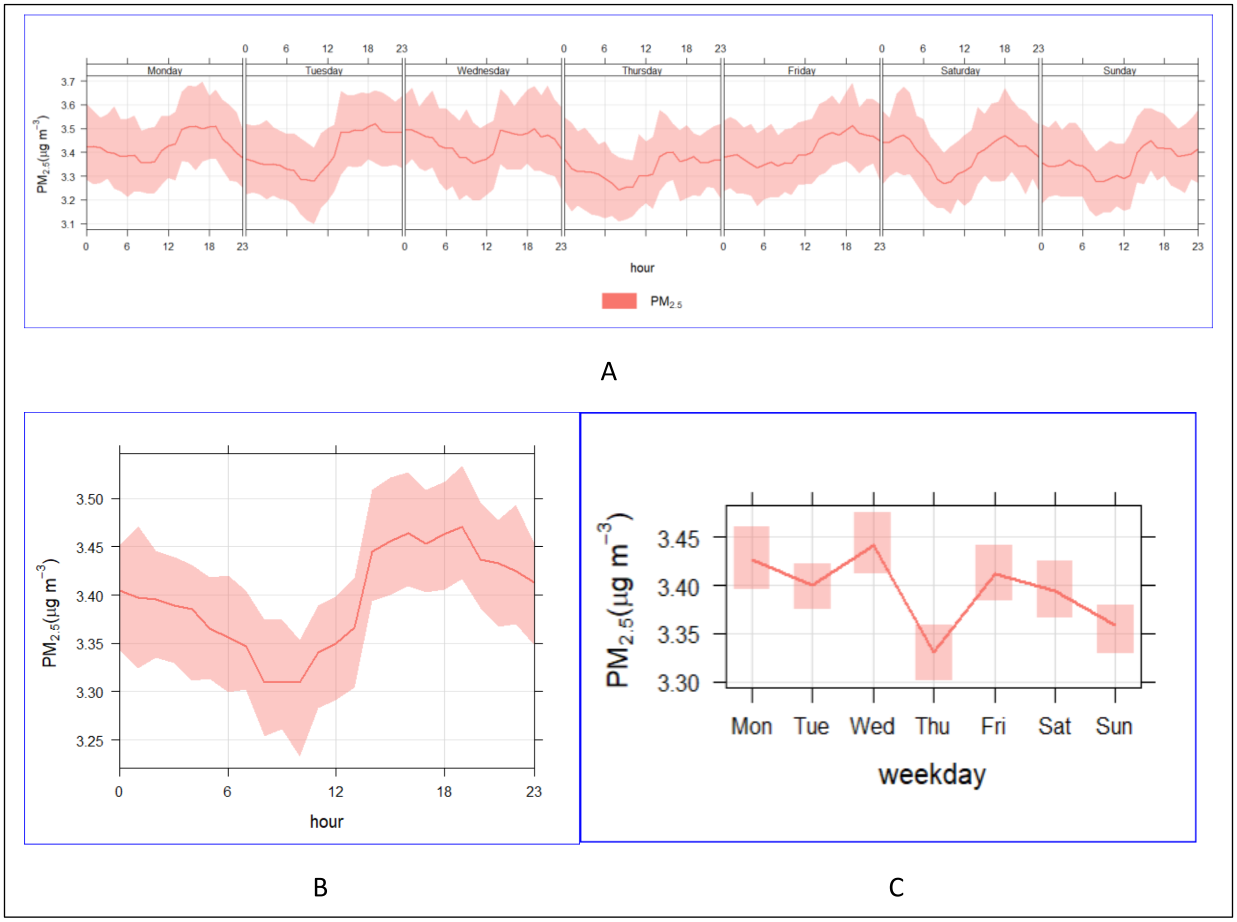

Figure 23: Diurnal variation of $PM_{2.5}$ in Anuradhapura

According to Figure 23, the hourly and weekly variation of $PM_{2.5}$ concentrations in Anuradhapura, Sri Lanka, reflects the city's climatic setting in the dry zone and its typical human activity patterns. The graphs show a consistent diurnal trend in $PM_{2.5}$ levels throughout the week, along with modest variations across weekdays.

On an hourly basis, $PM_{2.5}$ concentrations in Anuradhapura follow a clear diurnal cycle, with lower levels during the afternoon in local time and a gradual rise beginning around evening. This evening increase is likely related to enhanced local emissions from transportation, domestic activities, and minor industrial operations, combined with reduced vertical mixing due to rising temperatures that cap atmospheric dispersion. After sunset, the levels begin to decline slowly but stay relatively elevated compared to early mornings.

The weekly trend shows slight variations, with Monday and Friday registering slightly higher $PM_{2.5}$ levels, likely due to the start and end of the workweek when vehicular and human activity intensifies. Thursday appears to show the lowest concentrations, suggesting a relative dip in mid-week activity. Weekends (Saturday and Sunday) exhibit marginally lower $PM_{2.5}$ levels, consistent with reduced traffic and commercial activity in a city with less industrialization and a more residential character.

Jaffna

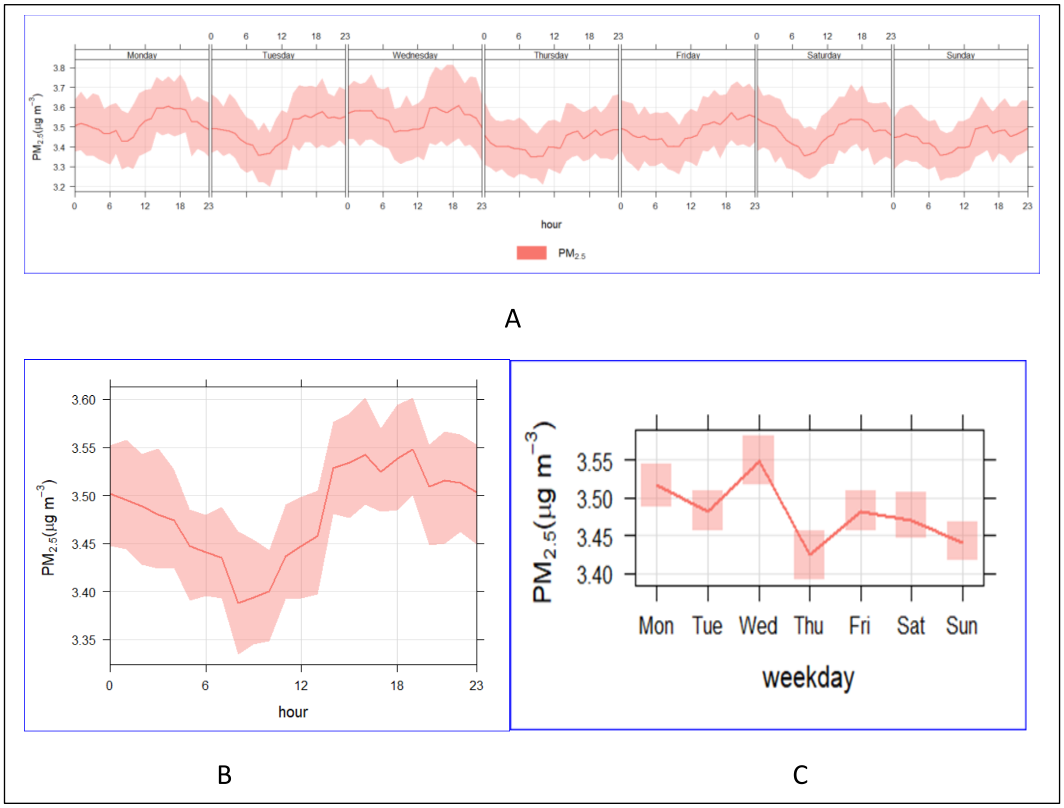

Figure 24: Diurnal variation of $PM_{2.5}$ in Jaffna

The diurnal and weekly patterns of $PM_{2.5}$ at Jaffna, Sri Lanka, as depicted in Figure 24, exhibit a consistently stable yet distinct trend that is controlled by the coastal climate characteristic of the region's dry zone and cycles of human activity. Diurnally, $PM_{2.5}$ concentrations are higher during the night in local time, with concentrations of approximately 3.55 µg m^{-3}, likely due to low wind speeds, cold air, and overnight accumulation of pollutants. Concentrations then decrease during the morning.

In terms of weekly variations, the levels of $PM_{2.5}$ in Jaffna display a high level of stability, with minimal fluctuations. On Wednesdays and Mondays, the highest average levels are witnessed, which might be attributed to increased commuting mid-week and post-weekend activities. Conversely, Thursdays and Sundays generally reflect relatively lower concentrations, reflecting reduced emissions these days because of either reduced traffic or commercial activities.

Comparative City Analysis

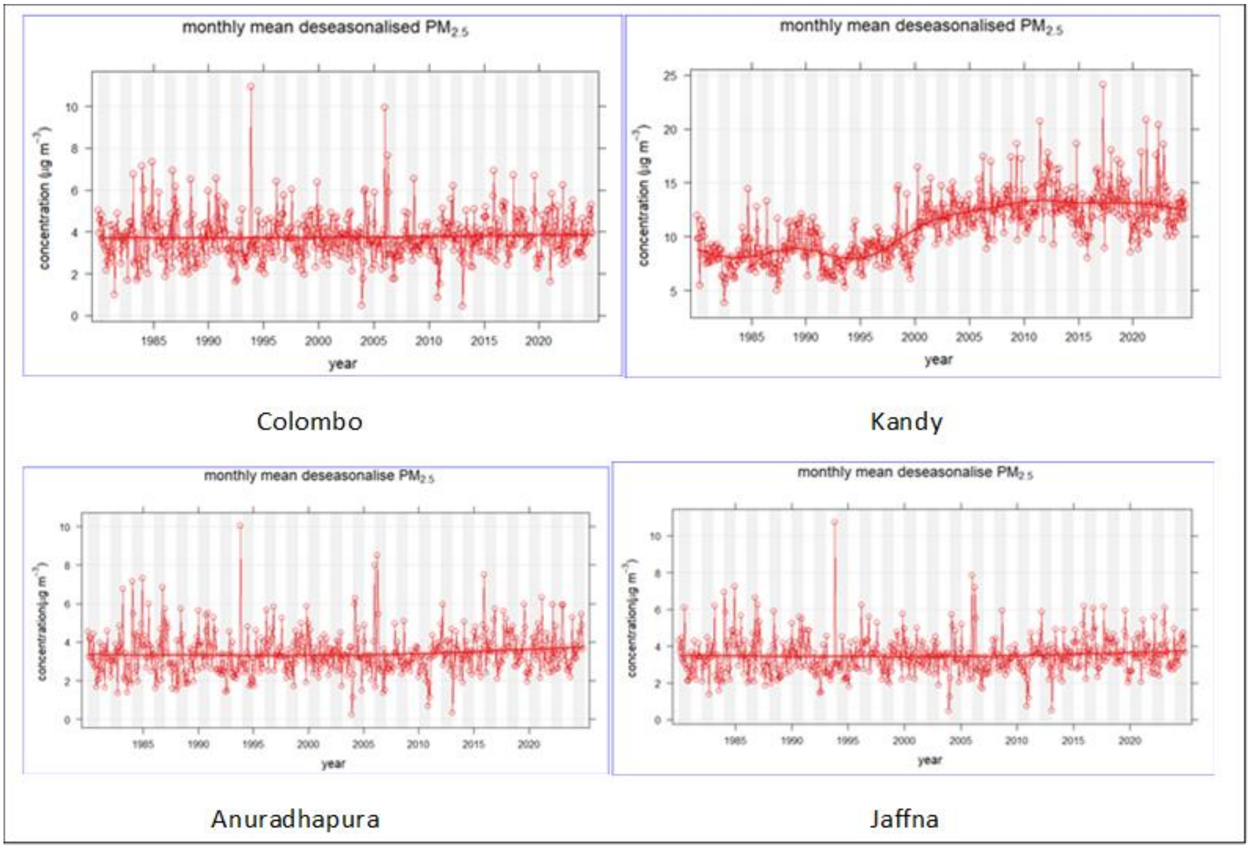

Figure 25: Monthly mean deseasonalise $PM_{2.5}$ in Colombo, Kandy, Anuradhapura, and Jaffna.

Figure 25 illustrates the deseasonalised monthly mean trends of $PM_{2.5}$ from 1980 to 2024 for the four studied Sri Lankan cities: Colombo, Kandy, Anuradhapura, and Jaffna. These trends show parallels and unique features that are influenced by local geography, patterns of urban growth, and atmospheric conditions.

Colombo, which is the commercial hub and the most urbanized city of the four investigated, shows a $PM_{2.5}$ level that is generally stable over the period of investigation. Trend analysis demonstrates the lack of pronounced increases or decreases; This suggests that while Colombo faces routine fluctuations, the general trend has been comparatively stable, which could be due to a sustained level of emissions being offset by coastal winds and introduced regulatory mechanisms.

In contrast, Kandy demonstrates increasing $PM_{2.5}$ levels since the mid-1990s. Beginning with quite low values during the 1980s, the deseasonalised mean increases steadily, reaching a plateau between the years 2015 and 2020. This increase is most likely a reflection of increasing urban density, road traffic, and limited dispersion because of the city's valley topography, which serves to trap pollutants. Of the four cities examined, Kandy exhibits the highest levels of $PM_{2.5}$ in recent years, which reflects a worsening problem of air quality.

Anuradhapura, being located in the dry zone with less industrial development, also exhibits the same pattern as Colombo. $PM_{2.5}$ concentrations are relatively stable. The general air quality pattern in the city is relatively stable and suggests lesser pressure from urban sources.

Similarly, the coastal city of Jaffna in the north shows little long-term trend in $PM_{2.5}$ concentrations. The trend is flat. The relatively low concentrations reflect that Jaffna is blessed with frequent coastal ventilation and less local sources of emissions.

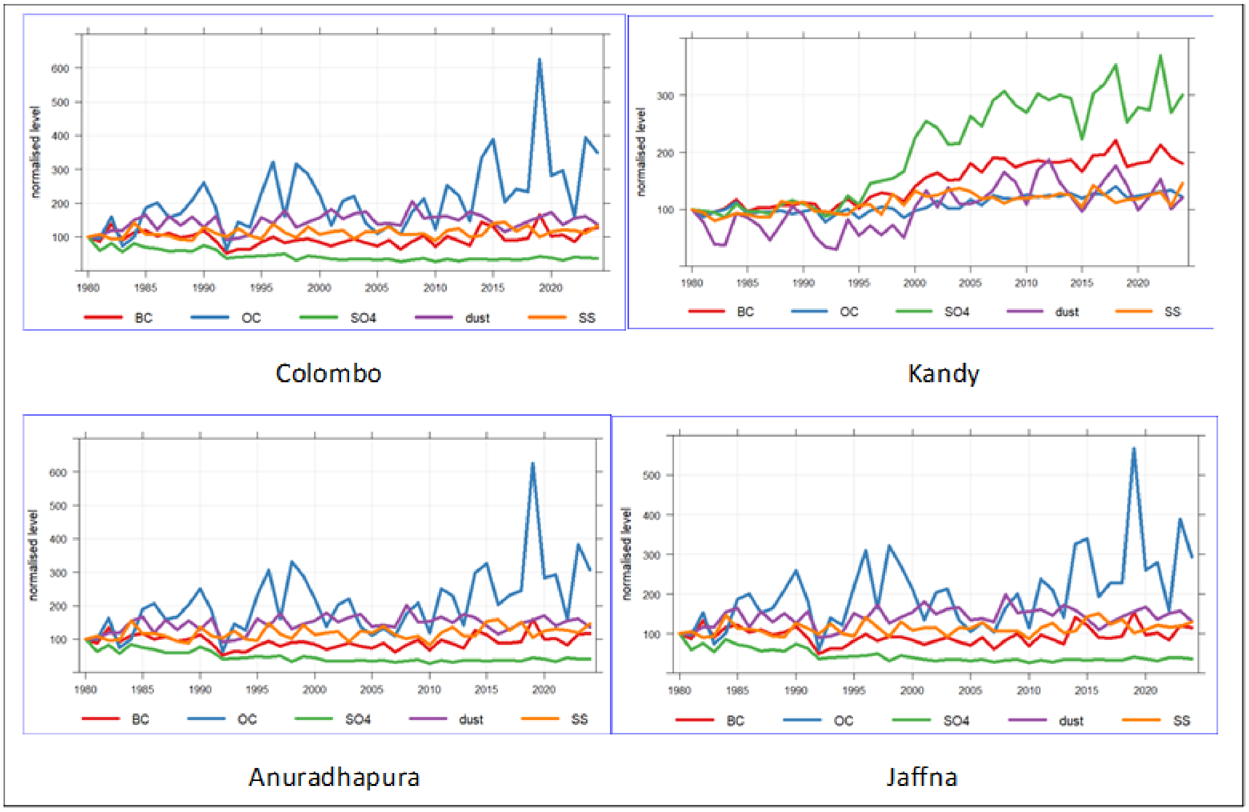

Figure 26: Normalising time series data in Colombo, Kandy, Anuradhapura and Jaffna to fix values to equal 100 at the beginning of 1980.

Figure 26 illustrates that in Colombo, OC exhibits a marked increase, particularly after 2000, with sharp peaks from 2018–2020, indicating heightened combustion-related activity, for example, traffic, industrial emissions, and diesel use. Other components, however, such as BC, SO_4, dust, and SS, are comparatively stable over time but with small increments. The observation implies that industrial and vehicular emission are becoming the dominant contributors towards Colombo's air pollution, especially in the recent past.

Kandy, in contrast, is characterized by a strong and steady SO_4 increase. The trend in OC is more constant. The modest increase in BC is observed, points to rising urban biomass burning, and vehicular emissions. The unique topography of Kandy (a basin surrounded by hills) likely allows pollutant trapping, which augments these trends. In comparison to other cities, SS and dust are steady, indicating less natural source impact.

At Anuradhapura, the trend is marked by rising OC, with episodic peaks, particularly in 2018–2020. The peaks resemble Colombo's trend but at somewhat lower levels. The remaining constituents are stable, with minimal variation in dust and SS. This is suggestive of low industrial activity but rising combustion emissions, perhaps due to vehicular growth or seasonal biomass burning in the dry zone.

Jaffna, a coastal city, also exhibits a similar profile to Anuradhapura and Colombo in the case of BC, with periodic variations being seen over the years. Jaffna also exhibits high concentrations of SS, in agreement with its maritime influence. The remaining constituents exhibit modest but persistent trends, suggesting minimal industrial emissions and the prevalence of natural aerosol contributions.

Except Kandy, other three cities, Colombo, Anuradhapura, and Jaffna shows similar results for $PM_{2.5}$ and its components during the study period.

Population Analysis

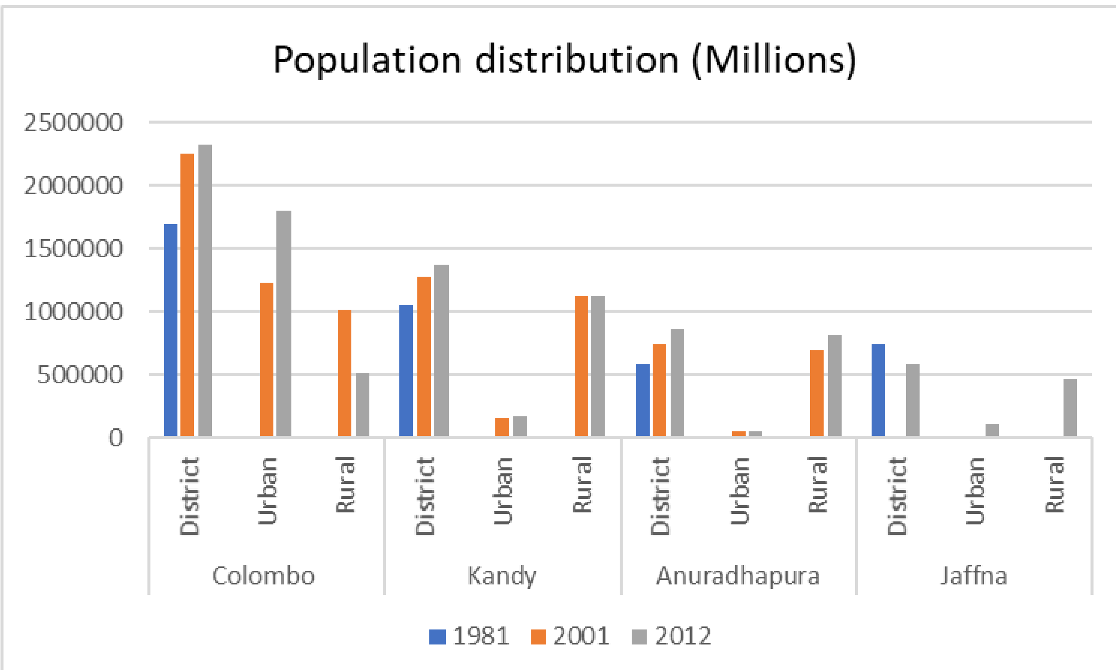

Figure 27: Population distribution in Colombo, Kandy, Anuradhapura, and Jaffna according to the district, urban, and rural

The chart shows the population pattern by district, urban, and rural areas of Colombo, Kandy, Anuradhapura, and Jaffna cities in 1981, 2001, and 2012. Figure 27 indicates that the population of the Colombo district has been the highest, with a consistent increase from 1981 to 2012. Meanwhile, the population of other urban centers is increasing in the same period, albeit at a lower rate. Surprisingly, the rural population in Colombo decreases by 2012, reflecting urbanization and possibly demonstrating that urban sprawl is extending into rural territories (DCS, 2001; DCS, 2015).

Kandy shows unequal distribution across urban, and rural population. The urban and rural categories show gradual growth between 1981 and 2012, which is reflective of the overall demographic growth in the urban as well as rural areas (DCS, 2001; DCS, 2015).

In Anuradhapura, the rural population is predominant, and both urban and rural exhibit consistent growth throughout the decades. This indicates continued growth in the rural population, perhaps because of agricultural lifestyles and a lower level of urban migration than in Colombo. The demographic trend in Jaffna cannot be described within the parameters of this thesis due to the lack of reliable statistics. In general, the graph displays the varying demographic patterns of the cities: urbanization in Colombo, balanced development in Kandy, rural dominance in Anuradhapura, and depopulation in Jaffna, according to varied economic, geographical, and historical explanations.

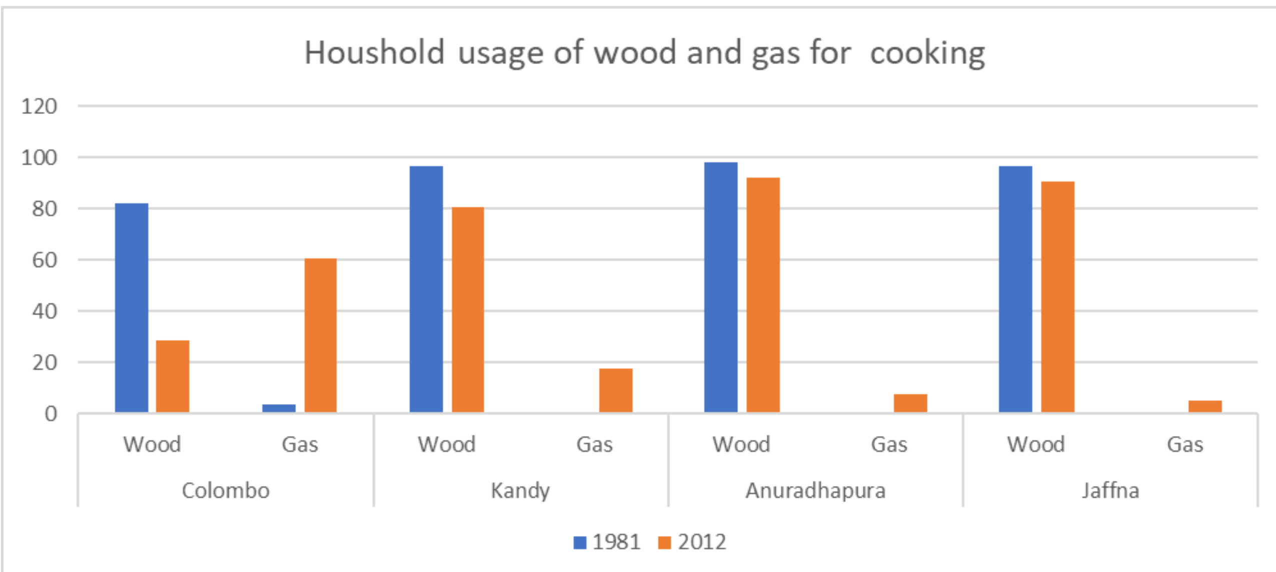

Figure 28: Household usage of wood and gas for cooking in Colombo, Kandy, Anuradhapura, and Jaffna during the years 1981 and 2012

In Colombo, there is a significant shift from wood to gas over time. In 1981, wood was the dominant cooking fuel, but by 2012, gas usage increased substantially, overtaking wood. This reflects urban modernization and greater access to cleaner energy in urban households (DCS, 2001; DCS, 2015).

Kandy shows a similar pattern, where gas usage more than doubled from 1981 to 2012, while wood usage decreased. Although wood was still widely used in 2012, the rise in gas suggests a growing adoption of cleaner cooking technologies (DCS, 2001; DCS, 2015).

In Anuradhapura, wood remains the primary cooking fuel even in 2012. Although gas usage slightly increased, the majority of households still rely on wood. This indicates limited access or slower transition to cleaner fuels in more rural or agrarian settings (DCS, 2001; DCS, 2015).

Same as Anuradhapura, Jaffna with minimal gas usage in both years and continued heavy reliance on wood, although there's a slight decrease by 2012. This trend may be linked to post-conflict recovery, infrastructure limitations, or economic constraints (DCS, 2001; DCS, 2015).

Overall, the graph reveals that urban areas like Colombo and Kandy are transitioning to gas, while Anuradhapura and Jaffna still rely heavily on wood, highlighting urban-rural disparities in energy access and fuel choices (DCS, 2001; DCS, 2015).

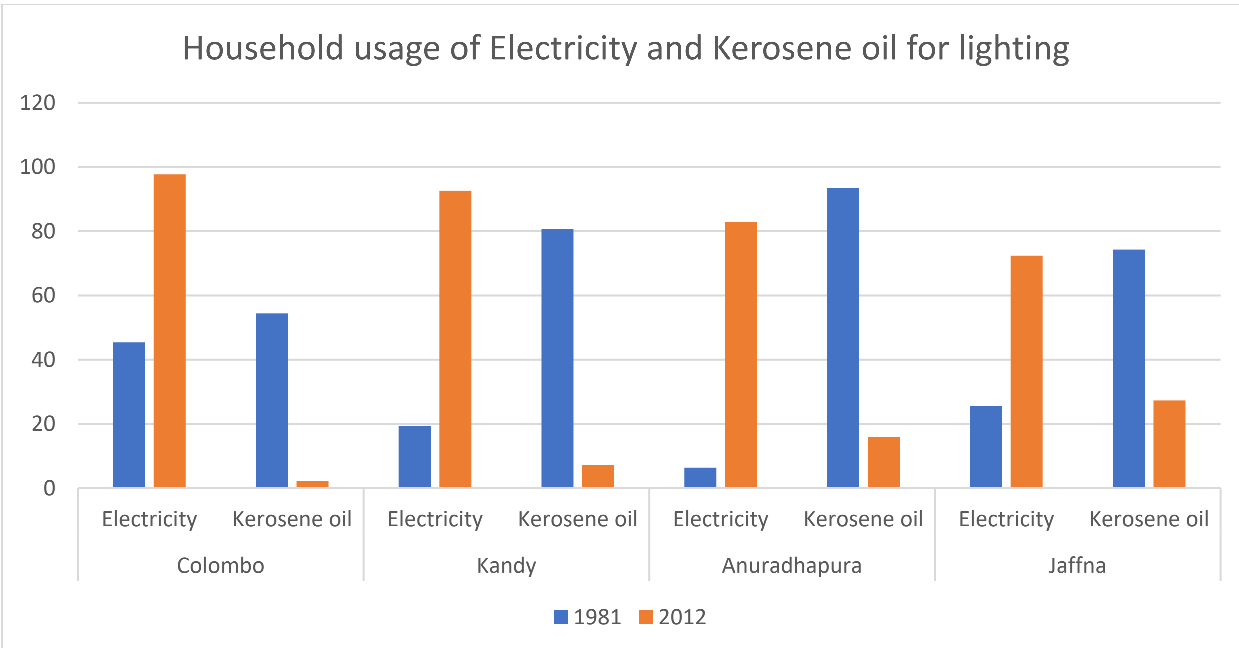

Figure 29: Household usage of electricity and kerosene oil for lighting in Colombo, Kandy, Anuradhapura, and Jaffna during the years 1981 and 2012

Figure 29 shows that household usage of electricity and kerosene oil for lighting in Colombo, Kandy, Anuradhapura, and Jaffna during the years 1981 and 2012. Further, due to the COVID-19 pandemic, plan census (2021) got postponed. Therefore, the latest data is not available. in Colombo, there was a notable increase in electricity usage for lighting from 1981 to 2012, while the use of kerosene oil dropped significantly. This suggests an urban transition toward more reliable and modern energy sources (DCS, 2001; DCS, 2015).

Similarly, Kandy experienced a sharp rise in electricity usage, reaching near-universal coverage by 2012. Kerosene oil, once widely used, became almost obsolete by this time, indicating strong infrastructure development and electrification (DCS, 2001; DCS, 2015).

In Anuradhapura, a more rural area, electricity usage also increased significantly over the three decades. However, kerosene oil remained relatively common in 1981, showing a slower transition compared to urban areas. By 2012, electricity overtook kerosene, demonstrating the spread of rural electrification programs (DCS, 2001; DCS, 2015).

In Jaffna, the transition was less pronounced. Although electricity usage increased, kerosene oil remained in use by a considerable portion of households in 2012. This could reflect regional disparities, post-conflict recovery, or infrastructure challenges. Overall, the graph reflects a nationwide shift from kerosene to electricity for lighting, particularly driven by urbanization and development, though regional gaps in access remain evident (DCS, 2001; DCS, 2015).

5. Discussion

Long-term Annual Trend Analysis

In this long-term analysis spanning over 45 years, with the exception of Kandy, the other three cities demonstrate a similar variation in $PM_{2.5}$, BC, OC, SO_4, dust, and SS. Before 2000, the concentration of $PM_{2.5}$ and its components showed low values, and after 2000, a small increase can be observed except SO_4. In Kandy, after 2000, it shows a significant increase in the concentration of each pollutant.

Kandy has undergone widespread urban expansion over the last decades, even if the population did not growth significantly. From 1996-2017, there has been a consistent decline in forest cover within the city, along with intensive road infrastructure and road development. This expansion has contributed to increased $PM_{2.5}$ and high interannual variability of dust concentrations. Figure 13 indicates that the southern region of Kandy experiences higher levels of $PM_{2.5}$, which is a condition most likely due to its greater population density and higher motor vehicle movement. These have been the causes of elevated levels of key pollutants like black carbon (BC), $PM_{2.5}$, and sulfate (SO_4) during the last several decades (Dissanayake et al., 2019).

As a response to rising levels of pollution, two fixed air quality monitoring stations were put up in Colombo in 2000 and 2003, targeting the effects of heightened vehicular and human activity (Seneviratne et al., 2011). All these initiatives, coupled with ever more comprehensive environmental policy, have guaranteed $PM_{2.5}$ stabilization in the area. Air quality regulations to safeguard public health have also been enforced by Sri Lanka. The acceptable national standard for $PM_{2.5}$ is 25 $\mu g\ m^{-3}$ on an annual average and 50 $\mu g\ m^{-3}$ on a 24-hour maximum basis (Seneviratne et al., 2011).

Trends in air quality have also been impacted by changes in the energy sector. In 1993, 95% of Sri Lanka's electricity was provided by hydropower. But in 2018, this proportion dipped to about 50%, with coal and thermal power largely taking its place. The increased reliance on coal-based power stations has contributed significantly to air pollution, particularly in the form of SO_2 emissions, which are sulfate aerosol precursors. Colombo, which houses over 50% of the nation's vehicles, 70% of industries, and some thermal power stations, has caused to change levels of $PM_{2.5}$, which is dominated by organic carbon (OC) emissions. Most notably, thermal power plants that utilize furnace oil containing approximately 3.5% sulfur content further boost SO_2 concentrations. However, the SO_4 concentration for Colombo did not show increase at all. This might be the policies, and other measures have been taken by the government.

The creation of the Central Environmental Authority (CEA) in 1981 was a milestone event in Sri Lanka's environmental administration. National ambient air quality standards were established in 1994, and the "Clean Air 2000 Action Plan" was launched in Colombo. The programme, under the Metropolitan Environmental Improvement Programme financed by the World Bank, sought to decrease urban air pollution by means of a mix of measures: lowering sulfur content in fuel, prohibiting two-stroke three-wheelers, eliminating leaded gasoline, and introducing vehicle emissions testing programmes (Ileperuma, 2020).

Seasonal Variation of $PM_{2.5}$ with Monsoon Patterns

Seasonal variations in $PM_{2.5}$ concentrations are most strongly associated with Sri Lanka's monsoonal climate. Apart from Kandy, $PM_{2.5}$ in Colombo, Anuradhapura, and Jaffna usually peaks in January and February, during the Northeast Monsoon (December to February). It is influenced by the semi-permanent Himalayan high-pressure system, which compels relatively low wind velocities over the island and transports polluted air masses from northern India.

An example of this was during November to January of 2019, when severe pollution in India had a significant contribution to the air quality in Sri Lanka. The transboundary influence was seen as peaks of BC, OC, and dust in Colombo, Anuradhapura, and Jaffna (Dhammapala et al., 2022; Ileperuma, 2020).

During the Southwest Monsoon (May to September), wind and rainfall patterns dramatically change.
Colombo and other western and southern cities receive maritime air and heavy rainfall, which spreads out pollutants and washes particles out of the air. As a result, $PM_{2.5}$ levels are lowest from June to August in Colombo, Anuradhapura, and Jaffna (Dhammapala et al., 2022; Ileperuma, 2020). This cleansing influence of the period illustrates the significant contribution of monsoonal circulation to the regulation of air pollution levels on the island.

Diurnal Variations of $PM_{2.5}$

The diurnal variation of $PM_{2.5}$ throughout Sri Lanka illustrates the usual urban activity patterns and domestic emissions. In the early morning, overnight-emitted pollutants are trapped near the surface by cooler temperatures and stable atmospheric conditions. As the sun rises, air temperature rises, bringing with it enhanced vertical mixing that enables the dispersal of pollutants.

When solar heating strengthens by noon in local time (7 +5.5 hours), dispersion in the vertical direction becomes better, however, the diurnal graphs do not show a clear decrease in $PM_{2.5}$ levels. This may due to local emissions remain high enough to offset dispersion effects. With cooling temperatures in the evening, atmospheric mixing is reduced again, and pollutants from domestic and traffic sources get trapped overnight. Biomass burning—both cooking and yard waste—is a significant source of $PM_{2.5}$, particularly in residential areas in Colombo.

either local emissions remain high enough to offset dispersion effects, or other factors—such as continuous traffic or cooking-related emissions

According to Figure 28, wood is still a prevalent cooking fuel in Kandy, Anuradhapura, and Jaffna, contributing to indoor and outdoor air pollution (Dhammapala et al., 2022).

Knowledge of diurnal behavior is essential to establish principal sources of pollution and assess the temporal dynamics of exposure risks. Such analysis guides effective policy responses in terms of both short-term advisories and long-term policy (Chérif et al., 2023). Comparative city analysis among the four researched locations, Kandy experiences the poorest air quality, with the highest long-term levels of $PM_{2.5}$ concentrations. This is mainly due to the fact that the city is basin-shaped, thus restricting air circulation and entrapping pollutants. It is also exacerbated by the rising number of vehicles, traffic congestion, and ongoing use of biomass fuels. Past records also indicate that SO_2 concentrations in Kandy surpassed country standards from 2002 to 2005, illustrating the impact of industrial and vehicular emissions on the pollution level. Over the last two decades, Kandy has also experienced an increase in respiratory illnesses, such as COPD, which may be attributed to worsening air quality. As evident from Figure 28, more than 70% of Kandy families continue to use wood as a significant fuel for cooking, which is a major contributor to the emission of $PM_{2.5}$ and other pollutants (Ileperuma, 2020). The reliance on solid fuels remains a significant hindrance to public health and environmental control within the region.

6. Conclusion

The present research analyzed the long-term trends, and seasonal and diurnal patterns, of $PM_{2.5}$ and its key components, BC, OC, SO_4, dust, and SS, in four principal cities of Sri Lanka: Colombo, Kandy, Anuradhapura, and Jaffna. Based on several data sources, such as MERRA-2 satellite data, Washington University datasets, and ground-level observations, the research identified major temporal and spatial patterns of pollution that were influenced by urbanization, topography, monsoonal seasons, and energy consumption.

Kandy registered the highest rise in $PM_{2.5}$ levels, particularly after 1996, with the increase mirroring increased urbanization, loss of forests, and expansion in transportation infrastructure. Colombo, which is more industrialized and densely urbanized, registered comparatively less variable $PM_{2.5}$ trends—possibly because of early policy intervention such as the Clean Air 2000 Action Plan and controls on vehicle exhaust.

Seasonal trends were seen in all cities. $PM_{2.5}$ concentrations peaking in the Northeast Monsoon months (December to February) and being at a minimum in the Southwest Monsoon months (May to September) in Colombo, Anuradhapura and Jaffna, highlighting the effect of transboundary pollution transport and local wind direction.

Overall, the study highlights population growth, energy transition, and policy adherence in shaping Sri Lanka's urban air quality. The study indicates the direction of more localized and integrated environmental management, particularly in physiographically vulnerable cities like Kandy.

7. Recommendations

Based on the findings of this research, the following are the recommendations suggested for enhancing urban air quality management and safeguarding public health in Sri Lanka:

1. Accelerate the Transition to Sustainable Home Energy Solutions

Increase access to electric and LPG cooking technology in regions where biomass use remains high, particularly Kandy, Anuradhapura, and Jaffna.

Implement financial incentives, subsidies, and sensitization campaigns for the use of clean alternative cooking fuels and to decrease indoor and outdoor air pollution exposure.

2. Enhance Urban Transport and Emissions Control

Increase public transportation facilities to reduce the use of private vehicles, especially at peak hours.

Enhance enforcement of motor vehicle emissions testing programs while phasing out high-emission vehicles, for example, two-stroke three-wheelers.

3. Enhance and Increase Air Quality Monitoring

Establish more fixed and mobile monitoring stations in underserved and high-risk areas.

Merge real-time satellite-based information with observations on the ground to improve forecasting and policy responsiveness.

4. Integrate Urban Planning with Air Quality Strategies

Include air quality assessments in urban planning development, especially in fast-growing cities such as Kandy.

Preserve current green areas and ventilation paths, and institute low-emission zonation policies within cities.

5. Address Regional and Seasonal Transport of Pollution

Establish cross-border mechanisms for early warning and collaborative response to transboundary air pollution, particularly during the Northeast Monsoon.

Integrate seasonal air quality forecasts into national environmental and public health planning.

6. Develop Public Awareness and Community Involvement.

Implement education programs at the community level regarding the health effects of air pollution and mitigation strategies.

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